
Measurement

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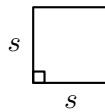
This chapter is a revision of perimeters and areas of two dimensional objects and volumes of three dimensional objects. We also examine different combinations of geometric objects and calculate areas and volumes in a variety of real-life contexts.

▶ See video: 2356 at www.everythingmaths.co.za

7.1 Area of a polygon

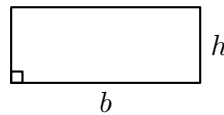
EMBHV

Square



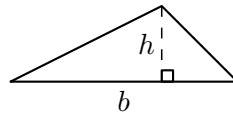
$$\text{Area} = s^2$$

Rectangle



$$\text{Area} = b \times h$$

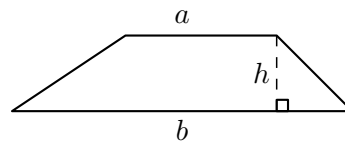
Triangle



$$\text{Area} = \frac{1}{2}b \times h$$

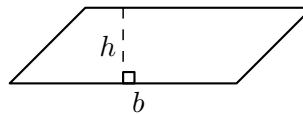
▶ See video: 2357 at www.everythingmaths.co.za

Trapezium



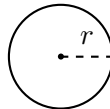
$$\text{Area} = \frac{1}{2}(a + b) \times h$$

Parallelogram



$$\text{Area} = b \times h$$

Circle



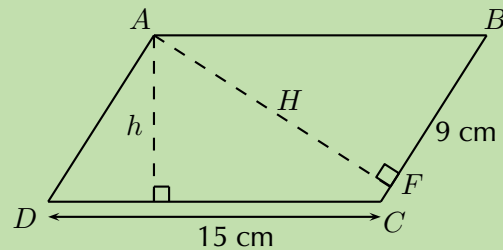
$$\begin{aligned} \text{Area} &= \pi r^2 \\ (\text{Circumference} &= 2\pi r) \end{aligned}$$

▶ See video: 2358 at www.everythingmaths.co.za

Worked example 1: Finding the area of a polygon

QUESTION

$ABCD$ is a parallelogram with $DC = 15$ cm, $h = 8$ cm and $BF = 9$ cm.



Calculate:

1. the area of $ABCD$
2. the perimeter of $ABCD$

SOLUTION

Step 1: Determine the area

The area of a parallelogram $ABCD = \text{base} \times \text{height}$:

$$\begin{aligned}\text{Area} &= 15 \times 8 \\ &= 120 \text{ cm}^2\end{aligned}$$

Step 2: Determine the perimeter

The perimeter of a parallelogram $ABCD = 2DC + 2BC$.

To find the length of BC , we use $AF \perp BC$ and the theorem of Pythagoras.

$$\begin{aligned}\text{In } \triangle ABF: \quad AF^2 &= AB^2 - BF^2 \\ &= 15^2 - 9^2 \\ &= 144 \\ \therefore AF &= 12 \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Area } ABCD &= BC \times AF \\ 120 &= BC \times 12 \\ \therefore BC &= 10 \text{ cm}\end{aligned}$$

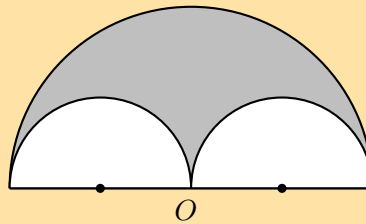
$$\begin{aligned}\therefore \text{Perimeter } ABCD &= 2(15) + 2(10) \\ &= 50 \text{ cm}\end{aligned}$$

Exercise 7 – 1: Area of a polygon

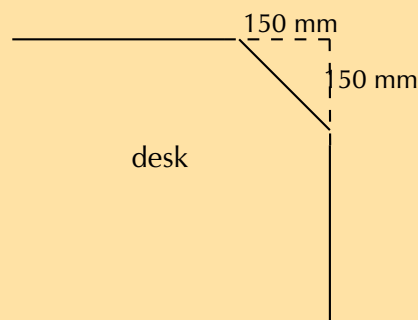
1. Vuyo and Banele are having a competition to see who can build the best kite using balsa wood (a lightweight wood) and paper. Vuyo decides to make his kite with one diagonal 1 m long and the other diagonal 60 cm long. The intersection of the two diagonals cuts the longer diagonal in the ratio 1 : 3.

Banele also uses diagonals of length 60 cm and 1 m, but he designs his kite to be rhombus-shaped.

- Draw a sketch of Vuyo's kite and write down all the known measurements.
 - Determine how much balsa wood Vuyo will need to build the outside frame of the kite (give answer correct to the nearest cm).
 - Calculate how much paper he will need to cover the frame of the kite.
 - Draw a sketch of Banele's kite and write down all the known measurements.
 - Determine how much wood and paper Banele will need for his kite.
 - Compare the two designs and comment on the similarities and differences. Which do you think is the better design? Motivate your answer.
2. O is the centre of the bigger semi-circle with a radius of 10 units. Two smaller semi-circles are inscribed into the bigger one, as shown on the diagram. Calculate the following (in terms of π):



- The area of the shaded figure.
 - The perimeter enclosing the shaded area.
3. Karen's engineering textbook is 30 cm long and 20 cm wide. She notices that the dimensions of her desk are in the same proportion as the dimensions of her textbook.
- If the desk is 90 cm wide, calculate the area of the top of the desk.
 - Karen uses some cardboard to cover each corner of her desk with an isosceles triangle, as shown in the diagram:



Calculate the new perimeter and area of the visible part of the top of her desk.

- c) Use this new area to calculate the dimensions of a square desk with the same desk top area.

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1. 2359 2. 235B 3. 235C



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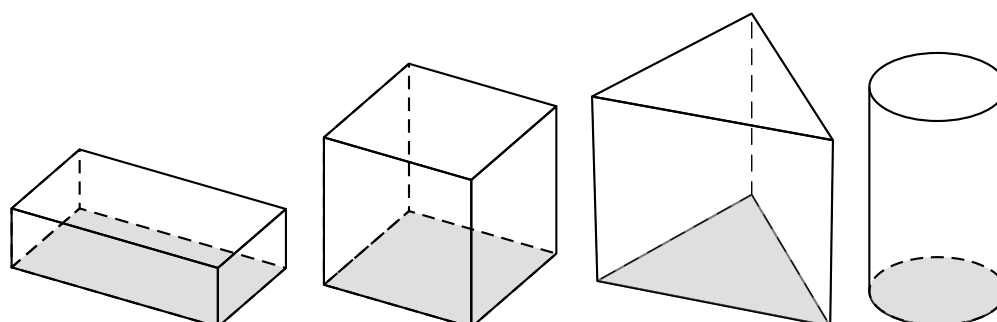
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7.2 Right prisms and cylinders

EMBHW

A right prism is a geometric solid that has a polygon as its base and vertical sides perpendicular to the base. The base and top surface are the same shape and size. It is called a “right” prism because the angles between the base and sides are right angles.

A triangular prism has a triangle as its base, a rectangular prism has a rectangle as its base, and a cube is a rectangular prism with all its sides of equal length. A cylinder is another type of right prism which has a circle as its base. Examples of right prisms are given below: a rectangular prism, a cube, a triangular prism and a cylinder.



Surface area of prisms and cylinders

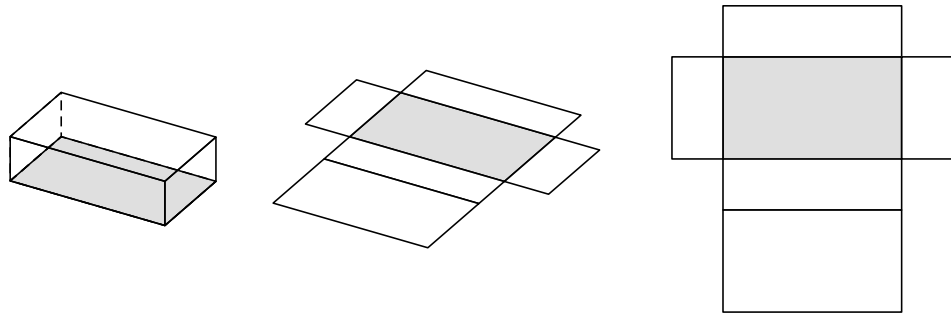
EMBHX

Surface area is the total area of the exposed or outer surfaces of a prism. This is easier to understand if we imagine the prism to be a cardboard box that we can unfold. A solid that is unfolded like this is called a net. When a prism is unfolded into a net, we can clearly see each of its faces. In order to calculate the surface area of the prism, we can then simply calculate the area of each face, and add them all together.

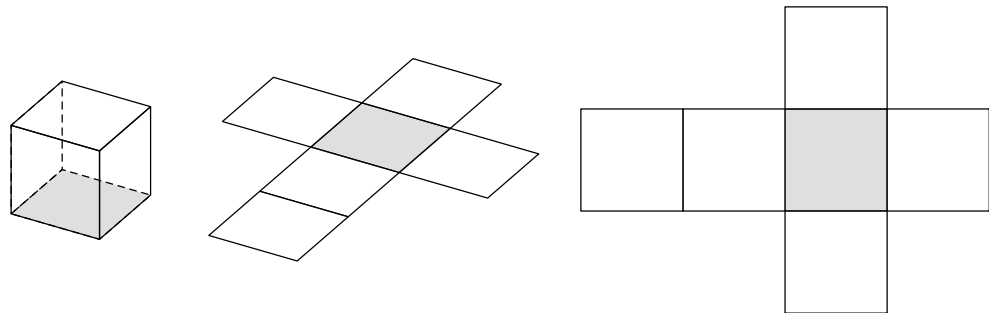
For example, when a triangular prism is unfolded into a net, we can see that it has two faces that are triangles and three faces that are rectangles. To calculate the surface area of the prism, we find the area of each triangle and each rectangle, and add them together.

In the case of a cylinder the top and bottom faces are circles and the curved surface flattens into a rectangle with a length that is equal to the circumference of the circular base. To calculate the surface area we therefore find the area of the two circles and the rectangle and add them together.

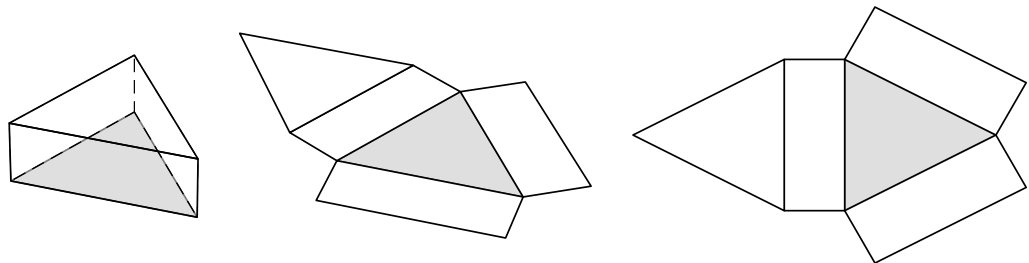
Below are examples of right prisms that have been unfolded into nets. A rectangular prism unfolded into a net is made up of six rectangles.



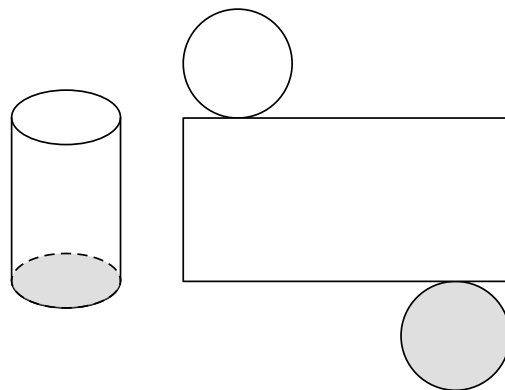
A cube unfolded into a net is made up of six identical squares.



A triangular prism unfolded into a net is made up of two triangles and three rectangles. The sum of the lengths of the rectangles is equal to the perimeter of the triangles.



A cylinder unfolded into a net is made up of two identical circles and a rectangle with length equal to the circumference of the circles.



Worked example 2: Calculating surface area

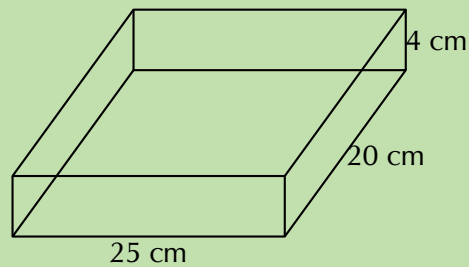
QUESTION

A box of chocolates has the following dimensions:

length = 25 cm

width = 20 cm

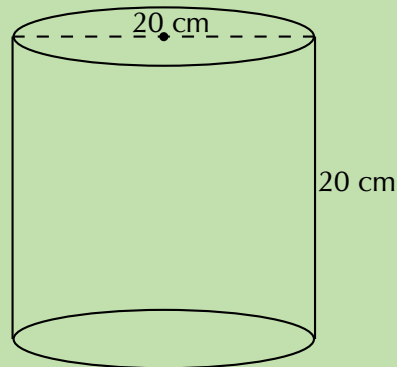
height = 4 cm



And a cylindrical tin of biscuits has the following dimensions:

diameter = 20 cm

height = 20 cm



1. Calculate the area of the wrapping paper needed to cover the entire box (assume no overlapping at the corners).
2. Determine if this same sheet of wrapping paper would be enough to cover the tin of biscuits.

SOLUTION

Step 1: Determine the area of the rectangular box

$$\begin{aligned}\text{Surface area} &= 2 \times (25 \times 20) + 2 \times (20 \times 4) + 2 \times (25 \times 4) \\ &= 1360 \text{ cm}^2\end{aligned}$$

Step 2: Determine the area of the cylindrical tin

The radius of the cylinder = $\frac{20}{2} = 10$ cm.

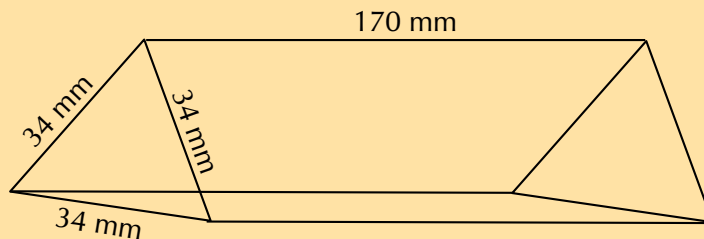
$$\begin{aligned}\text{Surface area} &= 2 \times \pi(10)^2 + 2\pi(10)(20) \\ &= 1885 \text{ cm}^2\end{aligned}$$

Step 3: Write the final answer

No, the area of the sheet of wrapping paper used to cover the box is not big enough to cover the tin.

Exercise 7 – 2: Calculating surface area

1. A popular chocolate container is an equilateral right triangular prism with sides of 34 mm. The box is 170 mm long. Calculate the surface area of the box (to the nearest square centimetre).

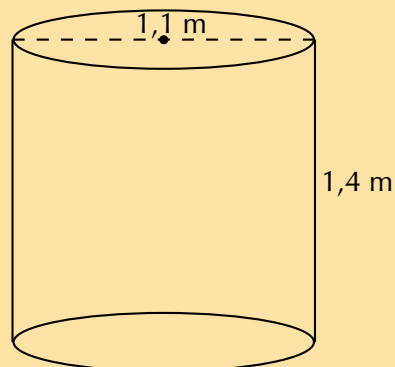


2. Gordon buys a cylindrical water tank to catch rain water off his roof. He discovers a full 2 ℓ tin of green paint in his garage and decides to paint the tank (not the base). If he uses 250 ml to cover 1 m², will he have enough green paint to cover the tank with one layer of paint?

Dimensions of the tank:

diameter = 1,1 m

height = 1,4 m



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1. 235D 2. 235F



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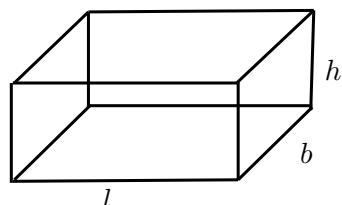


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Volume, sometimes also called capacity, is the three dimensional space occupied by an object, or the contents of an object. It is measured in cubic units.

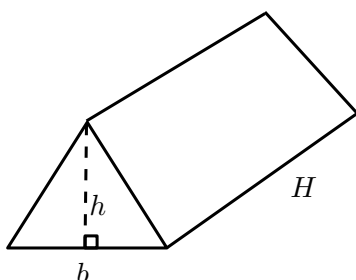
The volume of a right prism is simply calculated by multiplying the area of the base of a solid by the height of the solid.

Rectangular prism



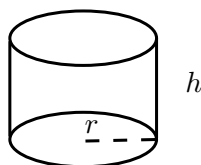
$$\begin{aligned}\text{Volume} &= \text{area of base} \times \text{height} \\ &= \text{area of rectangle} \times \text{height} \\ &= l \times b \times h\end{aligned}$$

Triangular prism



$$\begin{aligned}\text{Volume} &= \text{area of base} \times \text{height} \\ &= \text{area of triangle} \times \text{height} \\ &= \left(\frac{1}{2}b \times h\right) \times H\end{aligned}$$

Cylinder



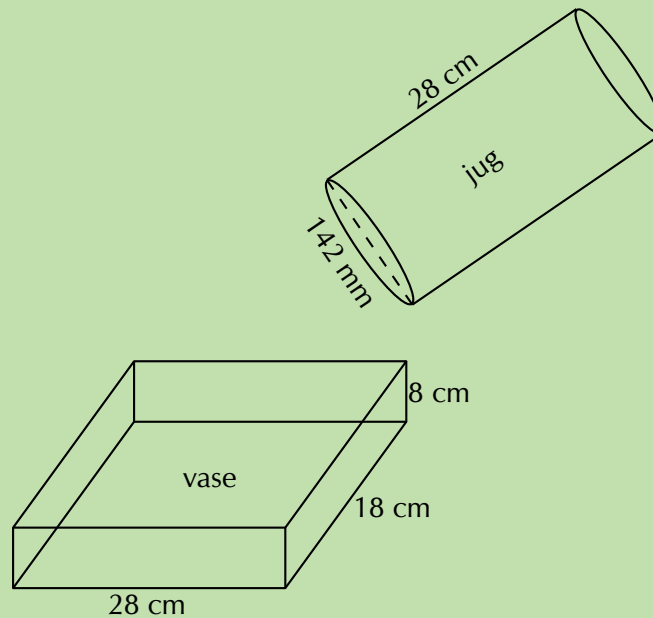
$$\begin{aligned}\text{Volume} &= \text{area of base} \times \text{height} \\ &= \text{area of circle} \times \text{height} \\ &= \pi r^2 \times h\end{aligned}$$

► See video: 235G at www.everythingmaths.co.za

Worked example 3: Calculating volume

QUESTION

A rectangular glass vase with dimensions $28\text{ cm} \times 18\text{ cm} \times 8\text{ cm}$ is used for flower arrangements. A florist uses a plastic cylindrical jug to pour water into the glass vase. The jug has a diameter of 142 mm and a height of 28 cm .



1. Will the plastic jug hold 5 l of water?
2. Will a full jug of water be enough to fill the glass vase?

SOLUTION

Step 1: Determine the volume of the plastic jug

The diameter of the jug is 142 mm , therefore the radius $= \frac{142}{2 \times 10} = 7,1\text{ cm}$.

Volume of a cylinder = area of the base $\times$ height

$$\begin{aligned}\text{Volume of the jug} &= \pi r^2 \times h \\ &= \pi \times (7,1)^2 \times 28 \\ &= 4434\text{ cm}^3\end{aligned}$$

$$\text{And } 1000\text{ cm}^3 = 1\text{ l}$$

$$\begin{aligned}\therefore \text{Volume of the jug} &= \frac{4434}{1000} \\ &= 4,434\text{ l}\end{aligned}$$

No, the capacity of the jug is not enough to hold 5 l of water.

Step 2: Determine the volume of the glass vase

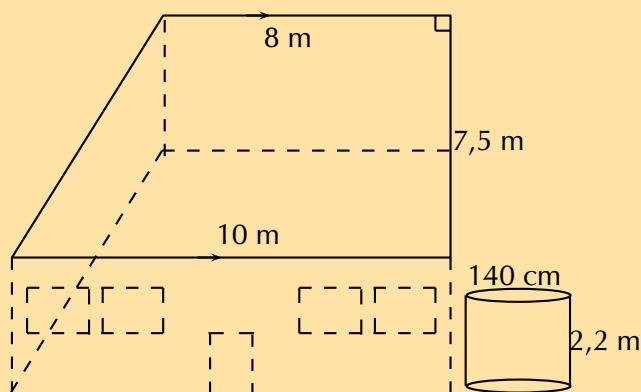
Volume of a rectangular prism = area of the base $\times$ height

$$\begin{aligned}\text{Volume of the vase} &= l \times b \times h \\ &= 28 \times 18 \times 8 \\ &= 4032 \text{ cm}^3 \\ \therefore \text{Volume of the vase} &= \frac{4032}{1000} \\ &= 4,032 \ell\end{aligned}$$

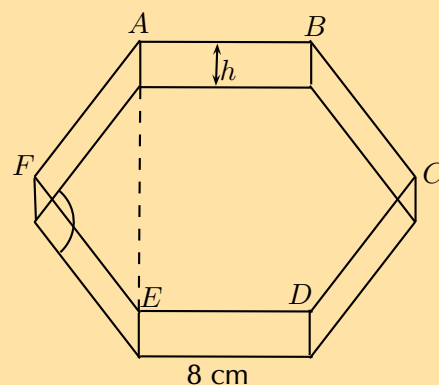
Yes, the volume of the jug is greater than the volume of the vase.

Exercise 7 – 3: Calculating volume

1. The roof of Phumza's house is the shape of a right-angled trapezium. A cylindrical water tank is positioned next to the house so that the rain on the roof runs into the tank. The diameter of the tank is 140 cm and the height is 2,2 m.



- a) Determine the area of the roof.
b) Determine how many litres of water the tank can hold.
2. The length of a side of a hexagonal sweet tin is 8 cm and its height is equal to half of the side length.



- a) Show that the interior angles are equal to 120° .

- b) Determine the length of the line AE .
- c) Calculate the volume of the tin.

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1. 235H 2. 235J



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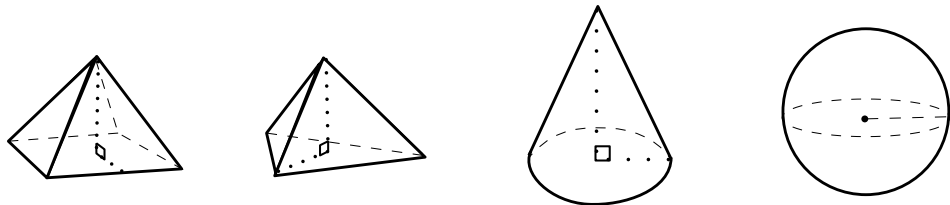


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7.3 Right pyramids, right cones and spheres

EMBJZ

A pyramid is a geometric solid that has a polygon as its base and sides that converge at a point called the apex. In other words the sides are **not** perpendicular to the base.

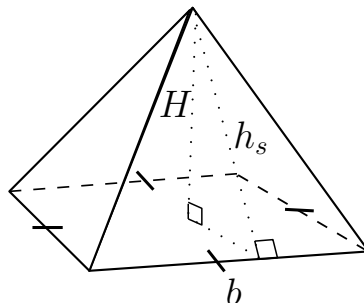


The triangular pyramid and square pyramid take their names from the shape of their base. We call a pyramid a “right pyramid” if the line between the apex and the centre of the base is perpendicular to the base. Cones are similar to pyramids except that their bases are circles instead of polygons. Spheres are solids that are perfectly round and look the same from any direction.

Surface area of pyramids, cones and spheres

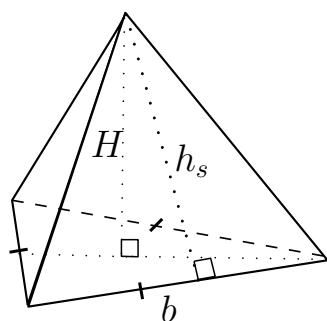
EMBJ2

Square pyramid



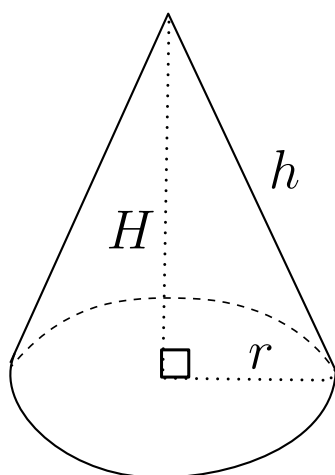
$$\begin{aligned}
 \text{Surface area} &= \text{area of base} + \text{area of triangular sides} \\
 &= b^2 + 4 \left(\frac{1}{2} b h_s \right) \\
 &= b(b + 2h_s)
 \end{aligned}$$

Triangular pyramid



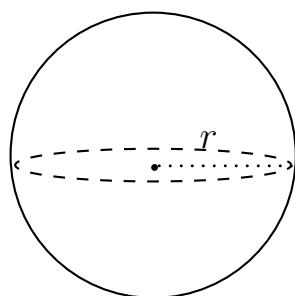
$$\begin{aligned}\text{Surface area} &= \text{area of base} + \\ &\quad \text{area of triangular sides} \\ &= \left(\frac{1}{2}b \times h_b\right) + 3\left(\frac{1}{2}b \times h_s\right) \\ &= \frac{1}{2}b(h_b + 3h_s)\end{aligned}$$

Right cone



$$\begin{aligned}\text{Surface area} &= \text{area of base} + \\ &\quad \text{area of walls} \\ &= \pi r^2 + \frac{1}{2} \times 2\pi r h \\ &= \pi r(r + h)\end{aligned}$$

Sphere

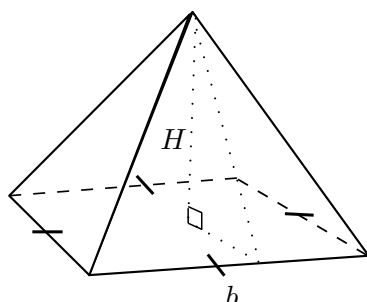


$$\text{Surface area} = 4\pi r^2$$

Volume of pyramids, cones and spheres

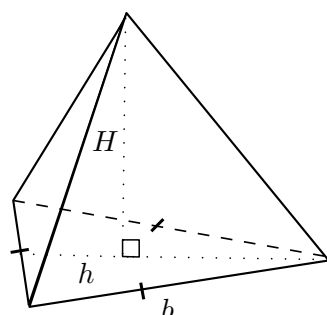
EMBJ3

Square pyramid



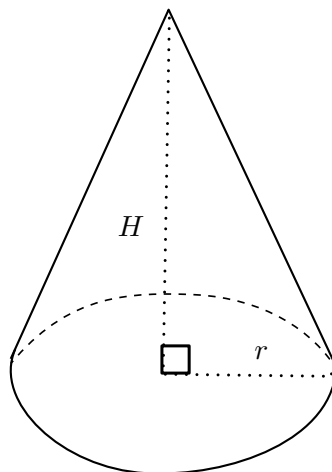
$$\begin{aligned}\text{Volume} &= \frac{1}{3} \times \text{area of base} \times \\ &\quad \text{height of pyramid} \\ &= \frac{1}{3} \times b^2 \times H\end{aligned}$$

Triangular pyramid



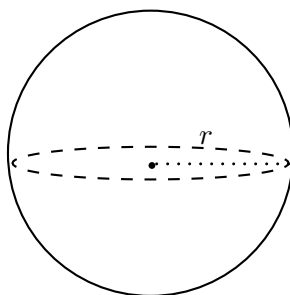
$$\begin{aligned}\text{Volume} &= \frac{1}{3} \times \text{area of base} \times \\ &\quad \text{height of pyramid} \\ &= \frac{1}{3} \times \frac{1}{2}bh \times H\end{aligned}$$

Right cone



$$\begin{aligned}\text{Volume} &= \frac{1}{3} \times \text{area of base} \times \text{height of cone} \\ &= \frac{1}{3} \times \pi r^2 \times H\end{aligned}$$

Sphere



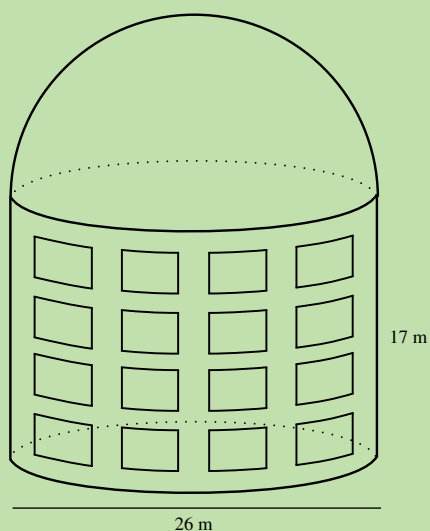
$$\text{Volume} = \frac{4}{3}\pi r^3$$

► See video: 235K at www.everythingmaths.co.za

Worked example 4: Finding surface area and volume

QUESTION

The Southern African Large Telescope (SALT) is housed in a cylindrical building with a domed roof in the shape of a hemisphere. The height of the building wall is 17 m and the diameter is 26 m.



1. Calculate the total surface area of the building.
2. Calculate the total volume of the building.

SOLUTION

Step 1: Calculate the total surface area

Total surface area = area of the dome + area of the cylinder

$$\begin{aligned}\text{Surface area} &= \left[\frac{1}{2}(4\pi r^2) \right] + [2\pi r \times h] \\ &= \frac{1}{2}(4\pi)(13)^2 + 2\pi(13)(17) \\ &= 2450 \text{ m}^2\end{aligned}$$

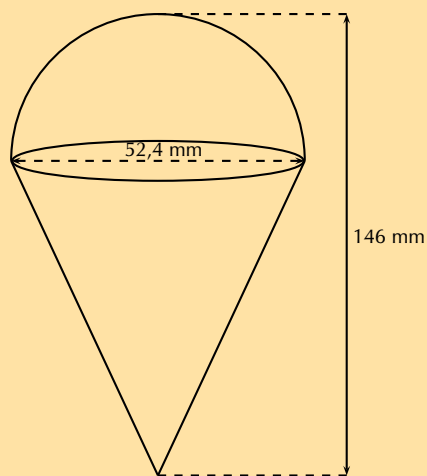
Step 2: Calculate the total volume

Total volume = volume of the dome + volume of the cylinder

$$\begin{aligned}\text{Volume} &= \left[\frac{1}{2} \times \left(\frac{4}{3}\pi r^3 \right) \right] + [\pi r^2 h] \\ &= \frac{2}{3}\pi(13)^3 + \pi(13)^2(17) \\ &= 9543 \text{ m}^3\end{aligned}$$

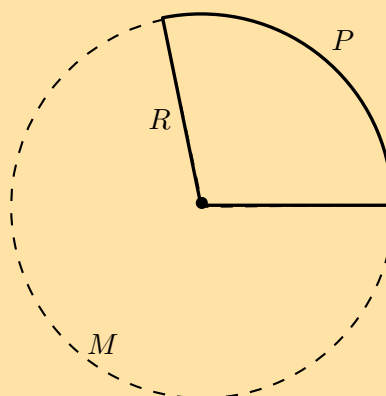
Exercise 7 – 4: Finding surface area and volume

1. An ice-cream cone has a diameter of 52,4 mm and a total height of 146 mm.



- Calculate the surface area of the ice-cream and the cone.
- Calculate the total volume of the ice-cream and the cone.
- How many ice-cream cones can be made from a 5 ℓ tub of ice-cream (assume the cone is completely filled with ice-cream)?

- d) Consider the net of the cone given below. R is the length from the tip of the cone to its perimeter, P .



- Determine the value of R .
- Calculate the length of arc P .
- Determine the length of arc M .

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1. 235M



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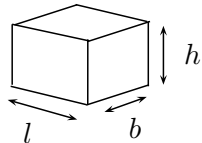
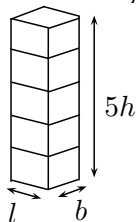
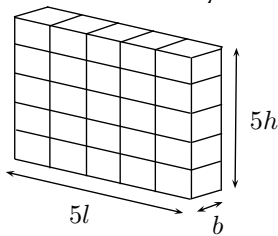
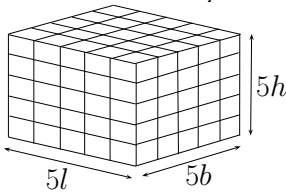
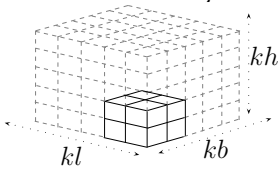
7.4 Multiplying a dimension by a constant factor

EMBJ4

When one or more of the dimensions of a prism or cylinder is multiplied by a constant, the surface area and volume will change. The new surface area and volume can be calculated by using the formulae from the preceding section.

It is important to see a relationship between the change in dimensions and the resulting change in surface area and volume. These relationships make it simpler to calculate the new volume or surface area of an object when its dimensions are scaled up or down.

Consider a rectangular prism of dimensions l , b and h . Below we multiply one, two and three of its dimensions by a constant factor of 5 and calculate the new volume and surface area.

Dimensions	Volume	Surface
Original dimensions 	$V = l \times b \times h$ $= lbh$	A $= 2[(l \times h) + (l \times b) + (b \times h)]$ $= 2(lh + lb + bh)$
Multiply one dimension by 5 	$V_1 = l \times b \times 5h$ $= 5(lbh)$ $= 5V$	A_1 $= 2[(l \times 5h) + (l \times b) + (b \times 5h)]$ $= 2(5lh + lb + 5bh)$
Multiply two dimensions by 5 	$V = 5l \times b \times 5h$ $= 5 \cdot 5(lbh)$ $= 5^2 V$	A_2 $= 2[(5l \times 5h) + (5l \times b) + (b \times 5h)]$ $= 2 \times 5(5lh + lb + bh)$
Multiply all three dimensions by 5 	$V = 5l \times 5b \times 5h$ $= 5^3(lbh)$ $= 5^3 V$	A_3 $= 2[(5l \times 5h) + (5l \times 5b) + (5b \times 5h)]$ $= 2 \times (5^2 lh + 5^2 lb + 5^2 bh)$ $= 5^2 \times 2(lh + lb + bh)$ $= 5^2 A$
Multiply all three dimensions by k 	$V = kl \times kb \times kh$ $= k^3(lbh)$ $= k^3 V$	A_k $= 2[(kl \times kh) + (kl \times kb) + (kb \times kh)]$ $= 2 \times (k^2 lh + k^2 lb + k^2 bh)$ $= k^2 \times 2(lh + lb + bh)$ $= k^2 A$

QUESTION

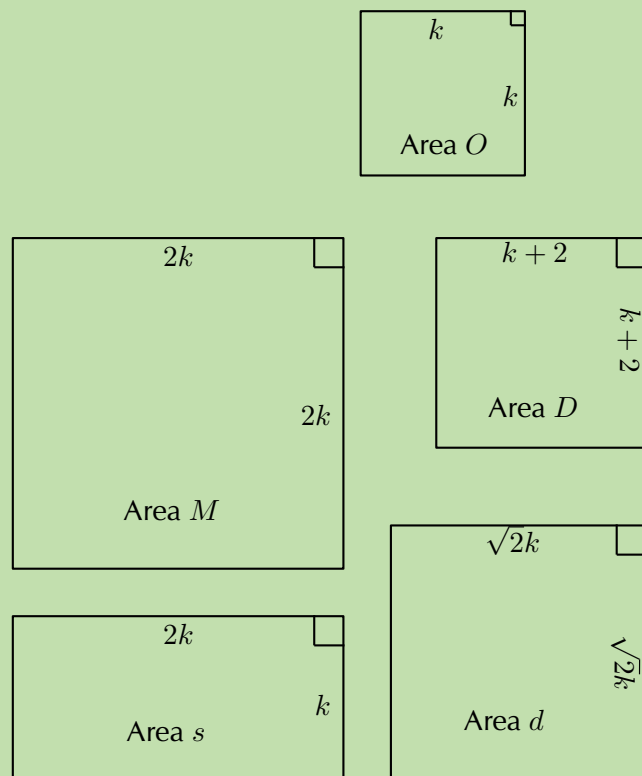
The Nash family wants to build a television room onto their house. The dad draws up the plans for the new square room of length k metres. The mum looks at the plans and decides that the area of the room needs to be doubled. To achieve this:

- the mum suggests doubling the length of the sides of the room
- the dad recommends adding 2 m to the length of the sides
- the daughter suggests multiplying the length of the sides by a factor of $\sqrt{2}$
- the son suggests doubling only the width of the room

Who's suggestion will double the area of the square room? Show all calculations.

SOLUTION

Step 1: Draw a sketch



Step 2: Calculate and compare

First calculate the area of the square room in the original plan:

$$\begin{aligned}\text{Area } O &= \text{length} \times \text{length} \\ &= k^2\end{aligned}$$

Therefore, double the area of the room would be $2k^2$.

Consider the mum's suggestion of doubling the length of the sides of the room:

$$\begin{aligned}\text{Area } M &= \text{length} \times \text{length} \\ &= 2k \times 2k \\ &= 4k^2\end{aligned}$$

This area would be 4 times the original area.

The dad suggests adding 2 m to the length of the sides of the room:

$$\begin{aligned}\text{Area } D &= \text{length} \times \text{length} \\ &= (k + 2) \times (k + 2) \\ &= k^2 + 4k + 2 \\ &\neq 2k^2\end{aligned}$$

This is not double the original area.

The daughter suggests multiplying the length of the sides by a factor of $\sqrt{2}$:

$$\begin{aligned}\text{Area } d &= \text{length} \times \text{length} \\ &= \sqrt{2}k \times \sqrt{2}k \\ &= 2k^2\end{aligned}$$

The daughter's suggestion would double the area of the room. Practically, the length of the room could be multiplied by $\sqrt{2} \approx 1,41$ which would give an area of $1,96 \text{ m}^2$.

The son suggests doubling only the width of the room:

$$\begin{aligned}\text{Area } s &= \text{length} \times \text{length} \\ &= 2k \times k \\ &= 2k^2\end{aligned}$$

The son's suggestion would double the area of the room, however the room would no longer be a square.

Step 3: Write the final answer

The daughter's suggestion of multiplying the length of the sides of the room by a factor of $\sqrt{2}$ would keep the shape of the room a square and would double the area of the room.

Exercise 7 – 5: The effects of k

1. Complete the following sentences:

- If one dimension of a cube is multiplied by a factor $\frac{1}{2}$, the volume of the cube ...
- If two dimensions of a cube are multiplied by a factor 7, the volume of the cube ...

- c) If three dimensions of a cube are multiplied by a factor 3, then:
- each side of the cube will ...
 - the outer surface area of the cube will ...
 - the volume of the cube will ...
- d) If each side of a cube is halved, then:
- the outer surface area of the cube will ...
 - the volume of the cube will ...
2. The municipality intends building a swimming pool of volume W^3 cubic metres. However, they realise that it will be very expensive to fill the pool with water, so they decide to make the pool smaller.
- The length and breadth of the pool are reduced by a factor of $\frac{7}{10}$. Express the new volume in terms of W .
 - The dimensions of the pool are reduced so that the volume of the pool decreases by a factor of 0,8. Determine the new dimensions of the pool in terms of W (remember that the pool must be a cube).

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1. 235N 2. 235P



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7.5 Summary

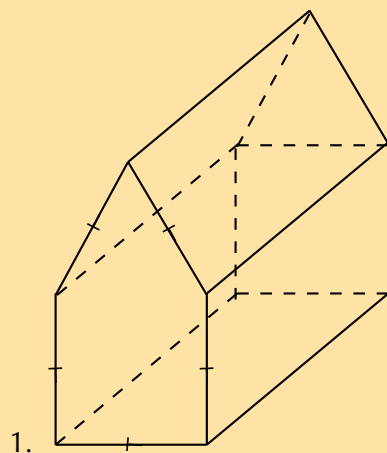
EMBJ5

▶ See presentation: 235Q at www.everythingmaths.co.za

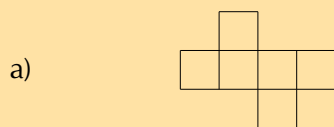
- Area is the two dimensional space inside the boundary of a flat object.
- Area formulae:
 - square: s^2
 - rectangle: $b \times h$
 - triangle: $\frac{1}{2}b \times h$
 - trapezium: $\frac{1}{2}(a + b) \times h$
 - parallelogram: $b \times h$
 - circle: πr^2
- Surface area is the total area of the exposed or outer surfaces of a prism.
- A net is the unfolded “plan” of a solid.
- Volume is the three dimensional space occupied by an object, or the contents of an object.
 - Volume of a rectangular prism: $l \times b \times h$

- Volume of a triangular prism: $(\frac{1}{2}b \times h) \times H$
 - Volume of a square prism or cube: s^3
 - Volume of a cylinder: $\pi r^2 \times h$
6. A pyramid is a geometric solid that has a polygon as its base and sides that converge at a point called the apex. The sides are not perpendicular to the base.
7. Surface area formulae:
- square pyramid: $b(b + 2h)$
 - triangular pyramid: $\frac{1}{2}b(h_b + 3h_s)$
 - right cone: $\pi r(r + h_s)$
 - sphere: $4\pi r^2$
8. Volume formulae:
- square pyramid: $\frac{1}{3} \times b^2 \times H$
 - triangular pyramid: $\frac{1}{3} \times \frac{1}{2}bh \times H$
 - right cone: $\frac{1}{3} \times \pi r^2 \times H$
 - sphere: $\frac{4}{3}\pi r^3$

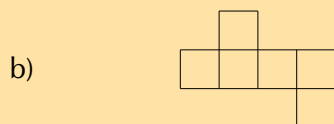
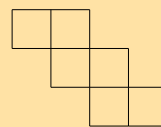
Exercise 7 – 6: End of chapter exercises



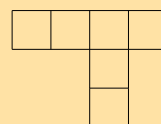
- a) Describe this figure in terms of a prism.
- b) Draw a net of this figure.
2. Which of the following is a net of a cube?



d)



e)

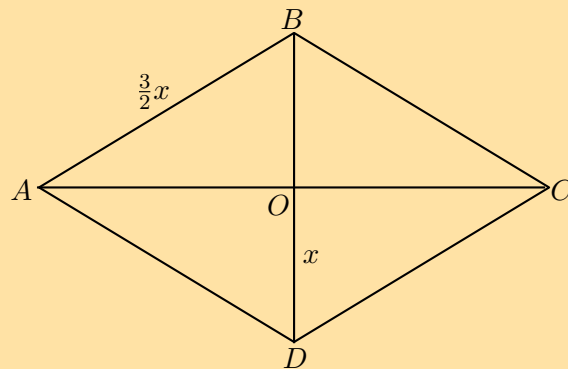


3. Name and draw the following figures:

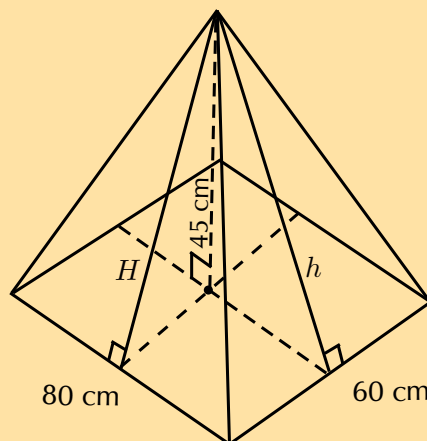
- a) A prism with the least number of sides.
- b) A pyramid with the least number of vertices.
- c) A right prism with a kite base.

4. a) i. Determine how much paper is needed to make a box of width 16 cm, height 3 cm and length 20 cm (assume no overlapping at corners).
 ii. Give a mathematical name for the shape of the box.
 iii. Calculate the volume of the box.
- b) Determine how much paper is needed to make a cube with a capacity of 1 ℓ.
- c) Compare the box and the cube. Which has the greater volume and which requires the most paper to make?

5. $ABCD$ is a rhombus with sides of length $\frac{3}{2}x$ millimetres. The diagonals intersect at O and length $DO = x$ millimetres. Express the area of $ABCD$ in terms of x .



6. The diagram shows a rectangular pyramid with a base of length 80 cm and breadth 60 cm. The vertical height of the pyramid is 45 cm.



- a) Calculate the volume of the pyramid.
 - b) Calculate H and h .
 - c) Calculate the surface area of the pyramid.
7. A group of children are playing soccer in a field. The soccer ball has a capacity of 5000 cc (cubic centimetres). A drain pipe in the corner of the field has a diameter of 20 cm. Is it possible for the children to lose their ball down the pipe? Show your calculations.

8. A litre of washing powder goes into a standard cubic container at the factory.
- Determine the length of the sides of the container.
 - Determine the dimensions of the cubic container required to hold double the volume of washing powder.
9. A cube has sides of length k units.
- Describe the effect on the volume of the cube if the height is tripled.
 - If all three dimensions of the cube are tripled, determine the effect on the outer surface area.
 - If all three dimensions of the cube are tripled, determine the effect on the volume.

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1. [235R](#) 2. [235S](#) 3a. [235T](#) 3b. [235V](#) 3c. [235W](#) 4. [235X](#)
5. [235Y](#) 6. [235Z](#) 7. [2362](#) 8. [2363](#) 9. [2364](#)



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Euclidean geometry

8.1	<i>Revision</i>	332
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8.1 Revision

EMBJ6

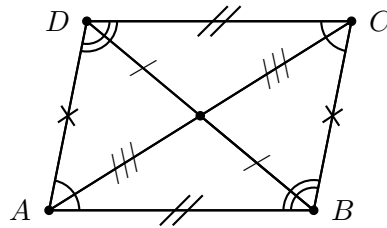
Parallelogram

EMBJ7

A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel.

Summary of the properties of a parallelogram:

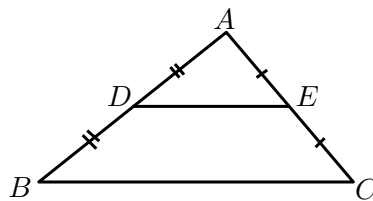
- Both pairs of opposite sides are parallel.
- Both pairs of opposite sides are equal in length.
- Both pairs of opposite angles are equal.
- Both diagonals bisect each other.



The mid-point theorem

EMBJ8

The line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.

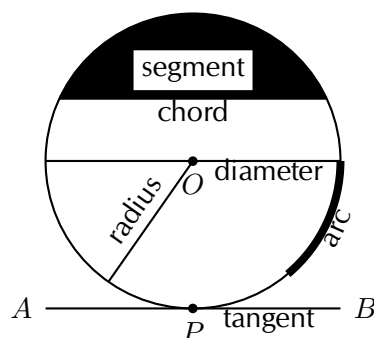


Given: $AD = DB$ and $AE = EC$, we can conclude that $DE \parallel BC$ and $DE = \frac{1}{2}BC$.

Terminology

The following terms are regularly used when referring to circles:

- **Arc** — a portion of the circumference of a circle.
- **Chord** — a straight line joining the ends of an arc.
- **Circumference** — the perimeter or boundary line of a circle.
- **Radius (r)** — any straight line from the centre of the circle to a point on the circumference.
- **Diameter** — a special chord that passes through the centre of the circle. A diameter is a straight line segment from one point on the circumference to another point on the circumference that passes through the centre of the circle.
- **Segment** — part of the circle that is cut off by a chord. A chord divides a circle into two segments.
- **Tangent** — a straight line that makes contact with a circle at only one point on the circumference.



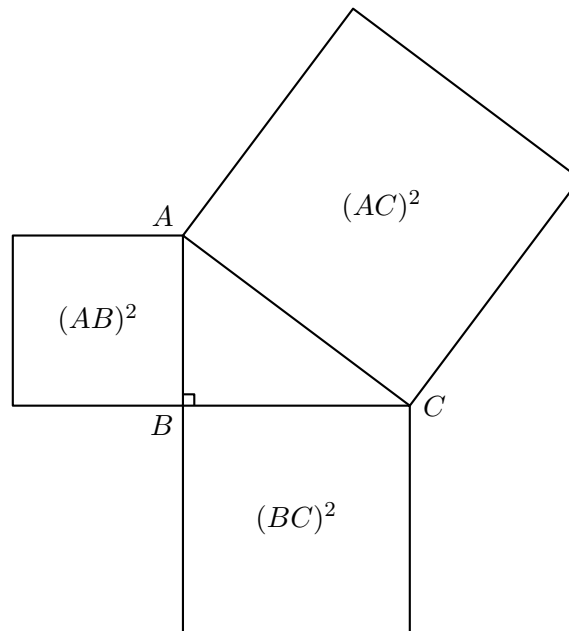
▶ See video: 2365 at www.everythingmaths.co.za

Axioms

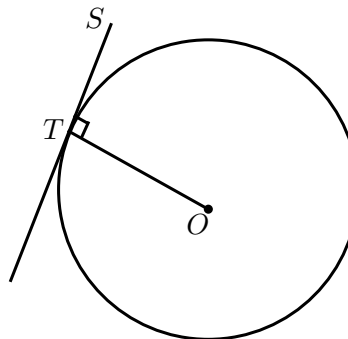
An axiom is an established or accepted principle. For this section, the following are accepted as axioms.

1. The theorem of Pythagoras states that the square of the hypotenuse of a right-angled triangle is equal to the sum of the squares of the other two sides.

$$(AC)^2 = (AB)^2 + (BC)^2$$



2. A tangent is perpendicular to the radius ($OT \perp ST$), drawn at the point of contact with the circle.



Theorems

EMBJB

A theorem is a hypothesis (proposition) that can be shown to be true by accepted mathematical operations and arguments. A proof is the process of showing a theorem to be correct.

The converse of a theorem is the reverse of the hypothesis and the conclusion. For example, given the theorem "if A , then B ", the converse is "if B , then A ".

Theorem: Perpendicular line from circle centre bisects chord

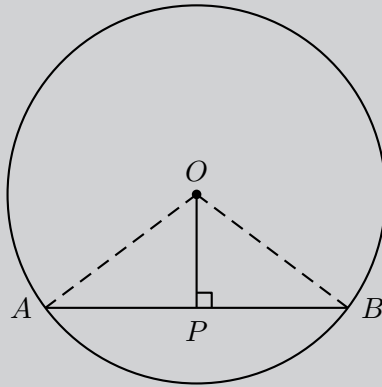
STATEMENT

If a line is drawn from the centre of a circle perpendicular to a chord, then it bisects the chord.

(Reason: $\perp$ from centre bisects chord)

Given:

Circle with centre O and line OP perpendicular to chord AB .



Required to prove:

$$AP = PB$$

PROOF

Draw OA and OB .

In $\triangle OPA$ and in $\triangle OPB$,

$$OA^2 = OP^2 + AP^2 \quad (\text{Pythagoras})$$

$$OB^2 = OP^2 + BP^2 \quad (\text{Pythagoras})$$

and

$$OA = OB \quad (\text{equal radii})$$

$$\therefore AP^2 = BP^2$$

$$\therefore AP = BP$$

Therefore OP bisects AB .

Alternative proof:

In $\triangle OPA$ and in $\triangle OPB$,

$$\angle OPA = \angle OPB \quad (\text{given } OP \perp AB)$$

$$OA = OB \quad (\text{equal radii})$$

$$OP = OP \quad (\text{common side})$$

$$\therefore \triangle OPA \equiv \triangle OPB \quad (\text{RHS})$$

$$\therefore AP = PB$$

Therefore OP bisects AB .

(PROOF NOT FOR EXAMS) Converse: Line from circle centre to mid-point of chord is perpendicular

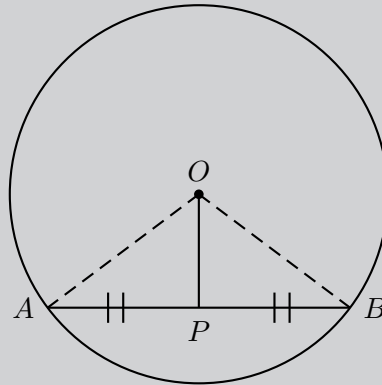
STATEMENT

If a line is drawn from the centre of a circle to the mid-point of a chord, then the line is perpendicular to the chord.

(Reason: line from centre to mid-point $\perp$)

Given:

Circle with centre O and line OP to mid-point P on chord AB .



Required to prove:

$OP \perp AB$

PROOF

Draw OA and OB .

In $\triangle OPA$ and in $\triangle OPB$,

$$OA = OB \quad (\text{equal radii})$$

$$AP = PB \quad (\text{given})$$

$$OP = OP \quad (\text{common side})$$

$$\therefore \triangle OPA \equiv \triangle OPB \quad (\text{SSS})$$

$$\therefore \hat{OPA} = \hat{OPB}$$

$$\text{and } \hat{OPA} + \hat{OPB} = 180^\circ \quad (\angle \text{ on str. line})$$

$$\therefore \hat{OPA} = \hat{OPB} = 90^\circ$$

Therefore $OP \perp AB$.

► See video: 2366 at www.everythingmaths.co.za

Theorem: Perpendicular bisector of chord passes through circle centre

STATEMENT

If the perpendicular bisector of a chord is drawn, then the line will pass through the centre of the circle.

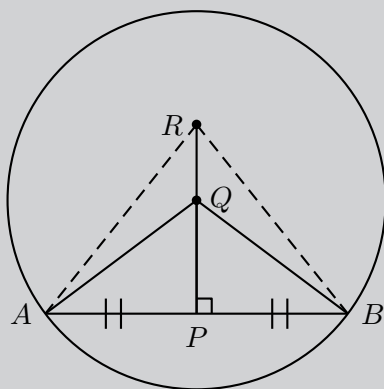
(Reason: $\perp$ bisector through centre)

Given:

Circle with mid-point P on chord AB .

Line QP is drawn such that $\hat{QPA} = \hat{QPB} = 90^\circ$.

Line RP is drawn such that $\hat{RPA} = \hat{RPB} = 90^\circ$.



Required to prove:

Circle centre O lies on the line PR

PROOF

Draw lines QA and QB .

Draw lines RA and RB .

In $\triangle QPA$ and in $\triangle QPB$,

$$\begin{array}{ll} AP = PB & \text{(given)} \\ QP = QP & \text{(common side)} \\ \hat{QPA} = \hat{QPB} = 90^\circ & \text{(given)} \\ \therefore \triangle QPA \equiv \triangle QPB & \text{(SAS)} \\ \therefore QA = QB & \end{array}$$

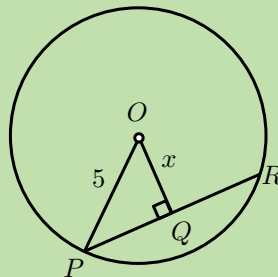
Similarly it can be shown that in $\triangle RPA$ and in $\triangle RPB$, $RA = RB$.

We conclude that all the points that are equidistant from A and B will lie on the line PR extended. Therefore the centre O , which is equidistant to all points on the circumference, must also lie on the line PR .

Worked example 1: Perpendicular line from circle centre bisects chord

QUESTION

Given $OQ \perp PR$ and $PR = 8$ units, determine the value of x .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles and sides on the diagram

$$PQ = QR = 4 \quad (\perp \text{ from centre bisects chord})$$

Step 2: Solve for x

In $\triangle OQP$:

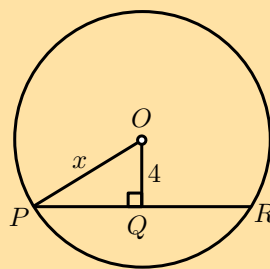
$$\begin{aligned} PQ &= 4 && (\perp \text{ from centre bisects chord}) \\ OP^2 &= OQ^2 + QP^2 && (\text{Pythagoras}) \\ 5^2 &= x^2 + 4^2 \\ \therefore x^2 &= 25 - 16 \\ x^2 &= 9 \\ x &= 3 \end{aligned}$$

Step 3: Write the final answer

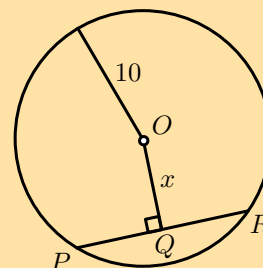
$x = 3$ units.

Exercise 8 – 1: Perpendicular line from center bisects chord

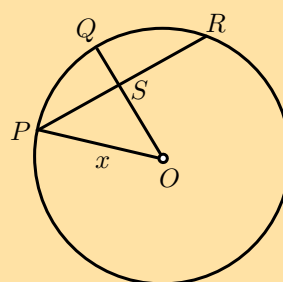
1. In the circle with centre O , $OQ \perp PR$, $OQ = 4$ units and $PR = 10$. Determine x .



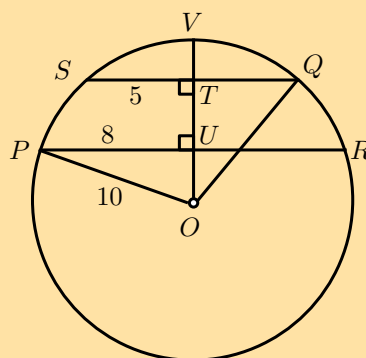
2. In the circle with centre O and radius = 10 units, $OQ \perp PR$ and $PR = 8$. Determine x .



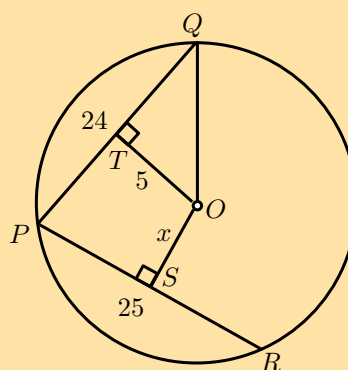
3. In the circle with centre O , $OQ \perp PR$, $PR = 12$ units and $SQ = 2$ units. Determine x .



4. In the circle with centre O , $OT \perp SQ$, $OT \perp PR$, $OP = 10$ units, $ST = 5$ units and $PU = 8$ units. Determine TU .



5. In the circle with centre O , $OT \perp QP$, $OS \perp PR$, $OT = 5$ units, $PQ = 24$ units and $PR = 25$ units. Determine $OS = x$.



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1. 2367 2. 2368 3. 2369 4. 236B 5. 236C



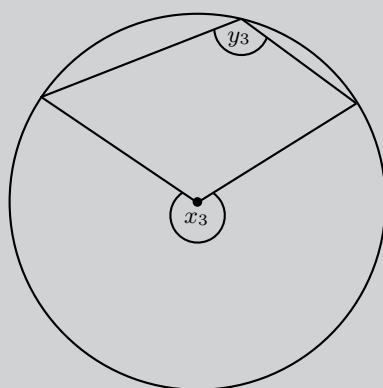
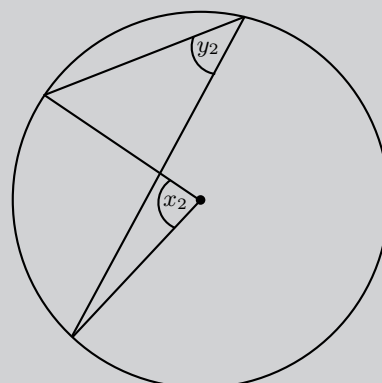
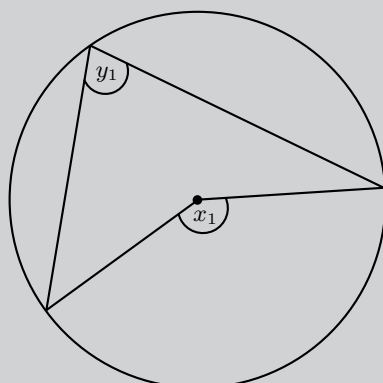
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Investigation: Angles subtended by an arc at the centre and the circumference of a circle

1. Measure angles x and y in each of the following graphs:



2. Complete the table:

x	y

3. Use your results to make a conjecture about the relationship between angles subtended by an arc at the centre of a circle and angles at the circumference of a circle.
4. Now draw three of your own similar diagrams and measure the angles to check your conjecture.

Theorem: Angle at the centre of a circle is twice the size of the angle at the circumference

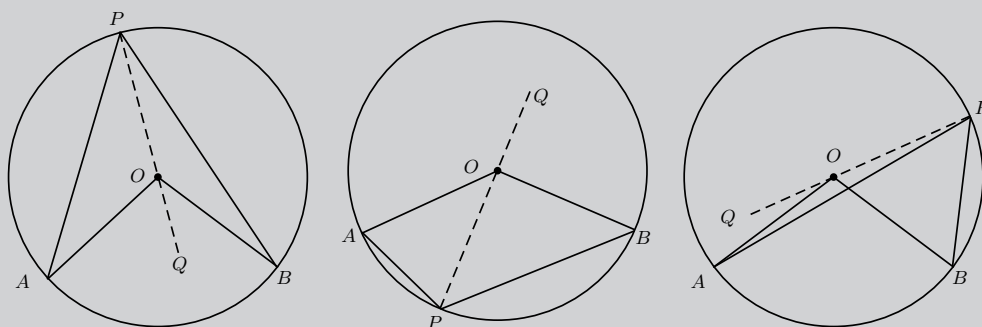
STATEMENT

If an arc subtends an angle at the centre of a circle and at the circumference, then the angle at the centre is twice the size of the angle at the circumference.

(Reason: $\angle$ at centre $= 2\angle$ at circum.)

Given:

Circle with centre O , arc AB subtending $\hat{AOB}$ at the centre of the circle, and $\hat{APB}$ at the circumference.



Required to prove:

$$\hat{AOB} = 2\hat{APB}$$

PROOF

Draw PO extended to Q and let $\hat{AOQ} = \hat{O}_1$ and $\hat{BOQ} = \hat{O}_2$.

$$\begin{aligned}\hat{O}_1 &= \hat{APO} + \hat{PAO} && (\text{ext. } \angle \triangle = \text{sum int. opp. } \angle \text{s}) \\ \text{and } \hat{APO} &= \hat{PAO} && (\text{equal radii, isosceles } \triangle APO) \\ \therefore \hat{O}_1 &= \hat{APO} + \hat{APO} \\ \hat{O}_1 &= 2\hat{APO}\end{aligned}$$

Similarly, we can also show that $\hat{O}_2 = 2\hat{BPO}$.

For the first two diagrams shown above we have that:

$$\begin{aligned}\hat{AOB} &= \hat{O}_1 + \hat{O}_2 \\ &= 2\hat{APO} + 2\hat{BPO} \\ &= 2(\hat{APO} + \hat{BPO}) \\ \therefore \hat{AOB} &= 2(\hat{APB})\end{aligned}$$

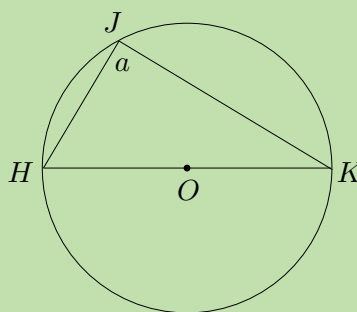
And for the last diagram:

$$\begin{aligned}\hat{AOB} &= \hat{O}_2 - \hat{O}_1 \\ &= 2\hat{BPO} - 2\hat{APO} \\ &= 2(\hat{BPO} - \hat{APO}) \\ \therefore \hat{AOB} &= 2(\hat{APB})\end{aligned}$$

Worked example 2: Angle at the centre of circle is twice angle at circumference

QUESTION

Given HK , the diameter of the circle passing through centre O .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles and sides on the diagram

Step 2: Solve for a

In $\triangle HJK$:

$$\begin{aligned} \hat{HOK} &= 180^\circ && (\angle \text{ on str. line}) \\ &= 2a && (\angle \text{ at centre} = 2\angle \text{ at circum.}) \\ \therefore 2a &= 180^\circ \\ a &= \frac{180^\circ}{2} \\ &= 90^\circ \end{aligned}$$

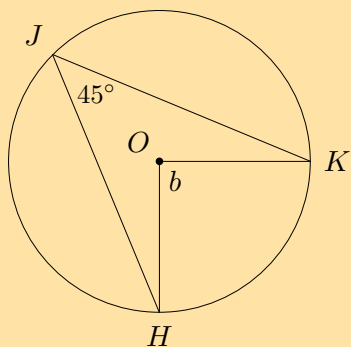
Step 3: Conclusion

The diameter of a circle subtends a right angle at the circumference (angles in a semi-circle).

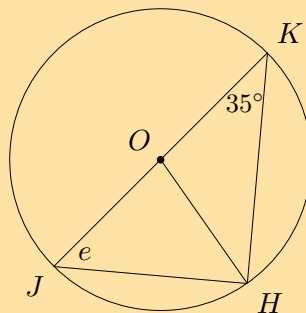
Exercise 8 – 2: Angle at the centre of circle is twice angle at circumference

Given O is the centre of the circle, determine the unknown angle in each of the following diagrams:

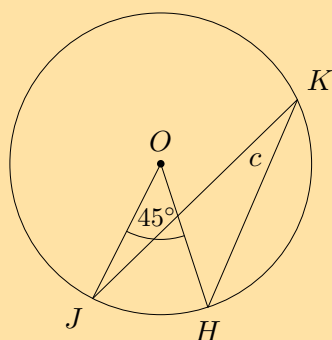
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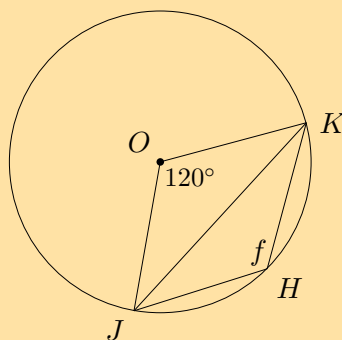
4.



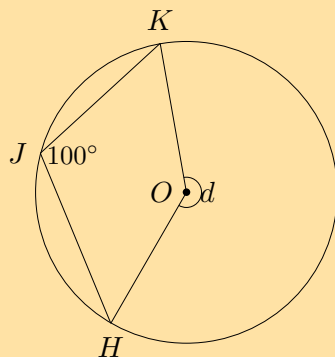
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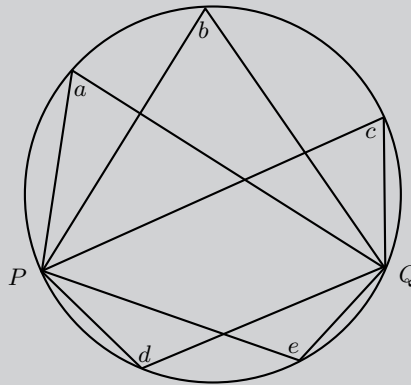
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Investigation: Subtended angles in the same segment of a circle

1. Measure angles a , b , c , d and e in the diagram below:



2. Choose any two points on the circumference of the circle and label them A and B .
3. Draw AP and BQ , and measure $\hat{APB}$.
4. Draw AQ and BQ , and measure $\hat{AQB}$.
5. What do you observe? Make a conjecture about these types of angles.

Theorem: Subtended angles in the same segment of a circle are equal

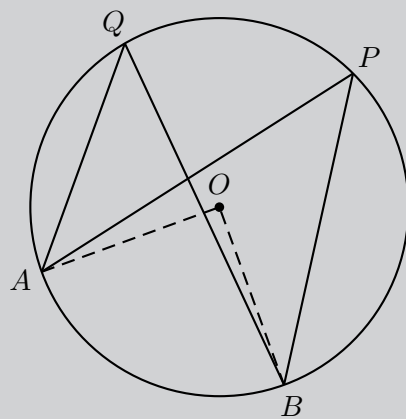
STATEMENT

If the angles subtended by a chord of the circle are on the same side of the chord, then the angles are equal.

(Reason: $\angle$ s in same seg.)

Given:

Circle with centre O , and points P and Q on the circumference of the circle. Arc AB subtends $\hat{APB}$ and $\hat{AQB}$ in the same segment of the circle.



Required to prove:

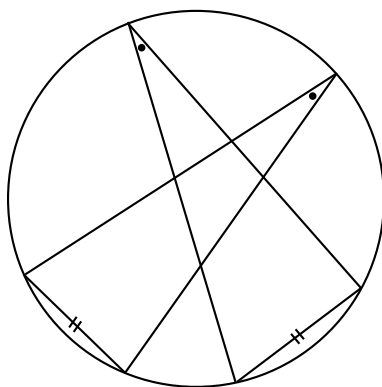
$$\angle APB = \angle AQB$$

PROOF

$$\begin{aligned} \angle AOB &= 2\angle APB & (\angle \text{ at centre} &= 2\angle \text{ at circum.}) \\ \angle AOB &= 2\angle AQB & (\angle \text{ at centre} &= 2\angle \text{ at circum.}) \\ \therefore 2\angle APB &= 2\angle AQB \\ \angle APB &= \angle AQB \end{aligned}$$

Equal arcs subtend equal angles

From the theorem above we can deduce that if angles at the circumference of a circle are subtended by arcs of equal length, then the angles are equal. In the figure below, notice that if we were to move the two chords with equal length closer to each other, until they overlap, we would have the same situation as with the theorem above. This shows that the angles subtended by arcs of equal length are also equal.

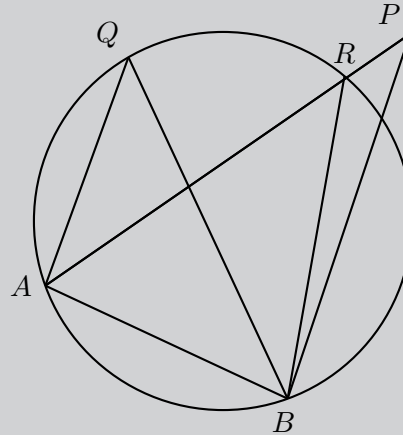


STATEMENT

If a line segment subtends equal angles at two other points on the same side of the line segment, then these four points are concyclic (lie on a circle).

Given:

Line segment AB subtending equal angles at points P and Q on the same side of the line segment AB .



Required to prove:

A, B, P and Q lie on a circle.

PROOF

Proof by contradiction:

Points on the circumference of a circle: we know that there are only two possible options regarding a given point — it either lies on circumference or it does not.

We will assume that point P does not lie on the circumference.

We draw a circle that cuts AP at R and passes through A, B and Q .

$$\begin{aligned} \angle AQB &= \angle ARB && (\angle\text{s in same seg.}) \\ \text{but } \angle AQB &= \angle APB && (\text{given}) \\ \therefore \angle ARB &= \angle APB \\ \text{but } \angle ARB &= \angle APB + \angle RBP && (\text{ext. } \angle\Delta = \text{sum int. opp.}) \\ \therefore \angle RBP &= 0^\circ \end{aligned}$$

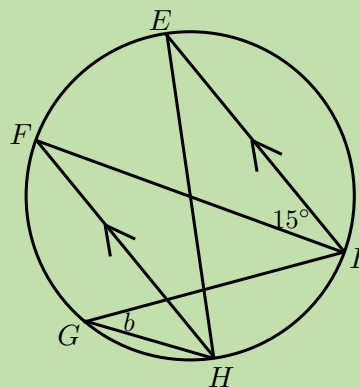
Therefore the assumption that the circle does not pass through P must be false.

We can conclude that A, B, Q and P lie on a circle (A, B, Q and P are concyclic).

Worked example 3: Concyclic points

QUESTION

Given $FH \parallel EI$ and $\hat{E}IF = 15^\circ$, determine the value of b .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles on the diagram

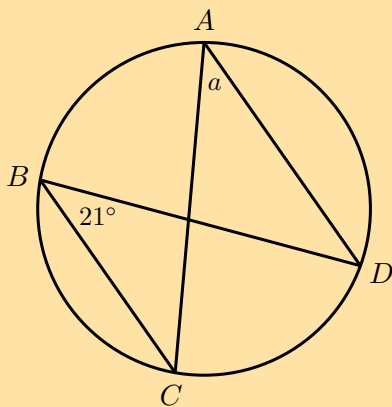
Step 2: Solve for b

$$\begin{aligned} \hat{HFI} &= 15^\circ && (\text{alt. } \angle, FH \parallel EI) \\ \text{and } b &= \hat{HFI} && (\angle\text{s in same seg.}) \\ \therefore b &= 15^\circ \end{aligned}$$

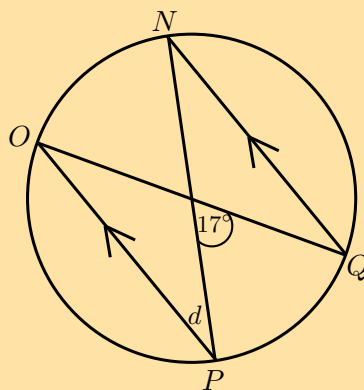
Exercise 8 – 3: Subtended angles in the same segment

1. Find the values of the unknown angles.

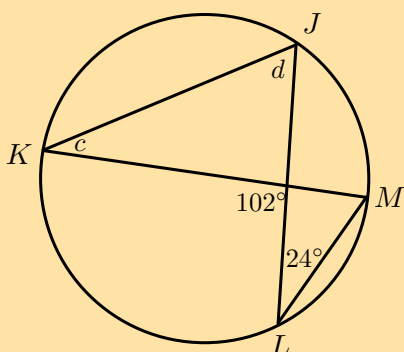
a)



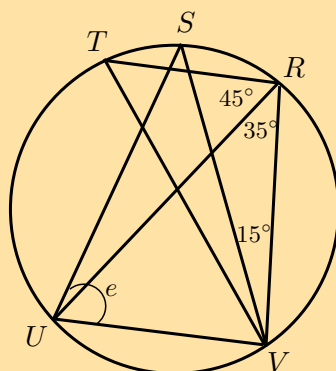
c)



b)

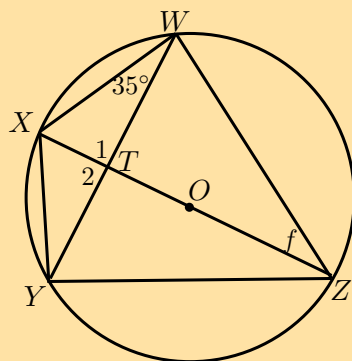


2.



- a) Given $\hat{TVS} = \hat{SVR}$, determine the value of e .
- b) Is TV a diameter of the circle? Explain your answer.

3.



Given circle with centre O , $WT = TY$ and $\hat{XWT} = 35^\circ$. Determine f .

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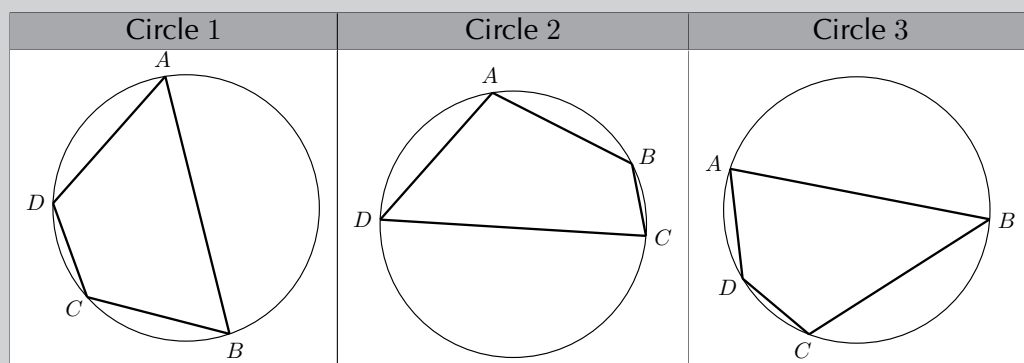
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Cyclic quadrilaterals

Cyclic quadrilaterals are quadrilaterals with all four vertices lying on the circumference of a circle (conyclic).

Investigation: Cyclic quadrilaterals

Consider the diagrams given below:



1. Complete the following:

$ABCD$ is a cyclic quadrilateral because

2. Complete the table:

	Circle 1	Circle 2	Circle 3
$\hat{A} =$			
$\hat{B} =$			
$\hat{C} =$			
$\hat{D} =$			
$\hat{A} + \hat{C} =$			
$\hat{B} + \hat{D} =$			

3. Use your results to make a conjecture about the relationship between angles of cyclic quadrilaterals.

Theorem: Opposite angles of a cyclic quadrilateral

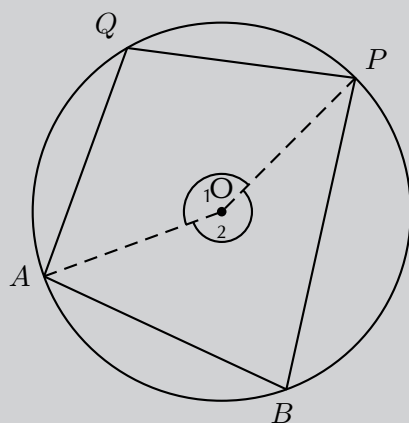
STATEMENT

The opposite angles of a cyclic quadrilateral are supplementary.

(Reason: opp. $\angle$ s cyclic quad.)

Given:

Circle with centre O with points A, B, P and Q on the circumference such that $ABPQ$ is a cyclic quadrilateral.



Required to prove:

$$\hat{A}BP + \hat{A}QP = 180^\circ \text{ and } \hat{Q}AB + \hat{Q}PB = 180^\circ$$

PROOF

Draw AO and OP . Label $\hat{O}_1$ and $\hat{O}_2$.

$$\begin{aligned}\hat{O}_1 &= 2\hat{A}\hat{B}P & (\angle \text{ at centre} = 2\angle \text{ at circum.}) \\ \hat{O}_2 &= 2\hat{A}\hat{Q}P & (\angle \text{ at centre} = 2\angle \text{ at circum.}) \\ \text{and } \hat{O}_1 + \hat{O}_2 &= 360^\circ & (\angle \text{ s around a point}) \\ \therefore 2\hat{A}\hat{B}P + 2\hat{A}\hat{Q}P &= 360^\circ \\ \hat{A}\hat{B}P + \hat{A}\hat{Q}P &= 180^\circ\end{aligned}$$

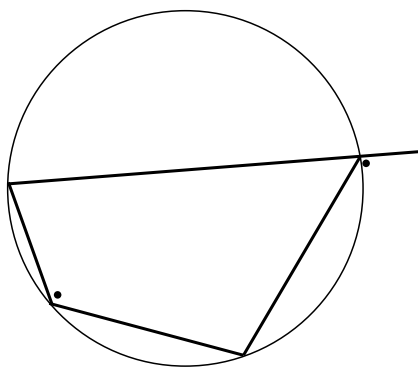
Similarly, we can show that $\hat{Q}\hat{A}B + \hat{Q}\hat{P}B = 180^\circ$.

Converse: interior opposite angles of a quadrilateral

If the interior opposite angles of a quadrilateral are supplementary, then the quadrilateral is cyclic.

Exterior angle of a cyclic quadrilateral

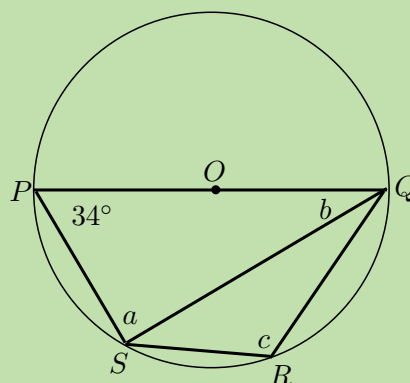
If a quadrilateral is cyclic, then the exterior angle is equal to the interior opposite angle.



Worked example 4: Opposite angles of a cyclic quadrilateral

QUESTION

Given the circle with centre O and cyclic quadrilateral $PQRS$. SQ is drawn and $\hat{S}\hat{P}Q = 34^\circ$. Determine the values of a , b and c .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles on the diagram

Step 2: Solve for b

$$S\hat{P}Q + c = 180^\circ \quad (\text{opp. } \angle\text{s cyclic quad suppl.})$$

$$\begin{aligned} \therefore c &= 180^\circ - 34^\circ \\ &= 146^\circ \end{aligned}$$

$$a = 90^\circ \quad (\angle \text{ in semi circle})$$

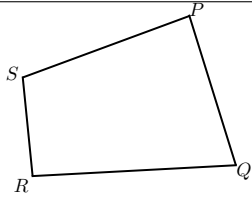
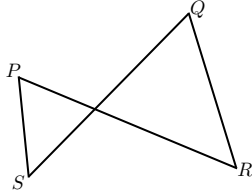
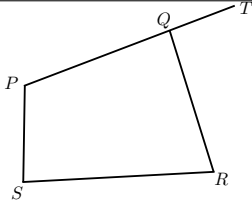
In $\triangle PSQ$:

$$a + b + 34^\circ = 180^\circ \quad (\angle \text{ sum of } \triangle)$$

$$\begin{aligned} \therefore b &= 180^\circ - 90^\circ - 34^\circ \\ &= 56^\circ \end{aligned}$$

Methods for proving a quadrilateral is cyclic

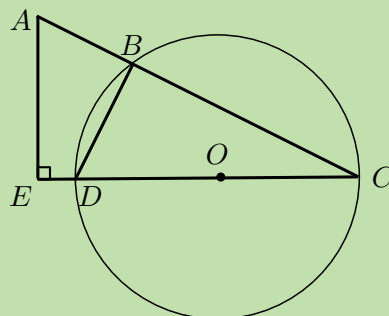
There are three ways to prove that a quadrilateral is a cyclic quadrilateral:

	Method of proof	Reason
	If $\hat{P} + \hat{R} = 180^\circ$ or $\hat{S} + \hat{Q} = 180^\circ$, then $PQRS$ is a cyclic quad.	opp. int. angles suppl.
	If $\hat{P} = \hat{Q}$ or $\hat{S} = \hat{R}$, then $PQRS$ is a cyclic quad.	angles in the same seg.
	If $T\hat{Q}R = \hat{S}$, then $PQRS$ is a cyclic quad.	ext. angle equal to int. opp. angle

Worked example 5: Proving a quadrilateral is a cyclic quadrilateral

QUESTION

Prove that $ABDE$ is a cyclic quadrilateral.



SOLUTION

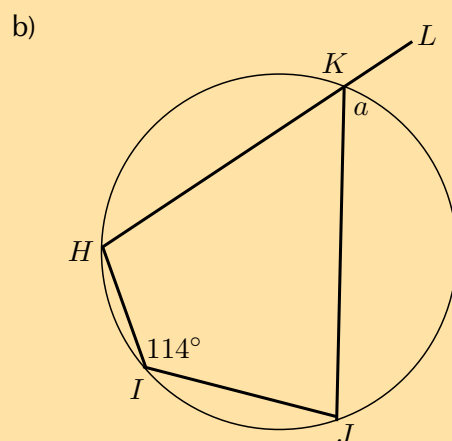
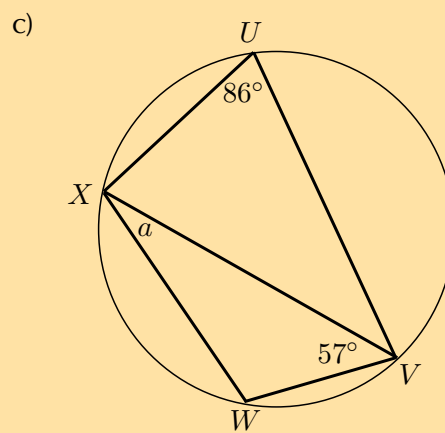
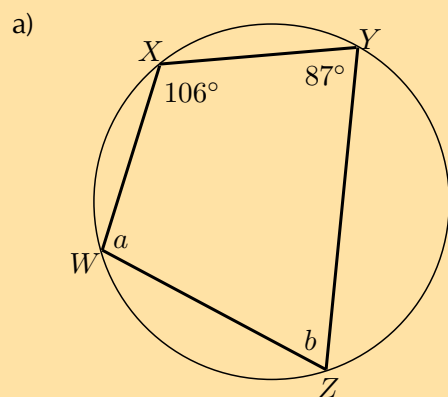
Step 1: Use theorems and the given information to find all equal angles on the diagram

Step 2: Prove that $ABDE$ is a cyclic quadrilateral

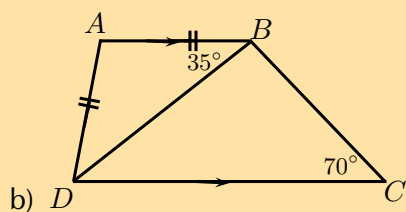
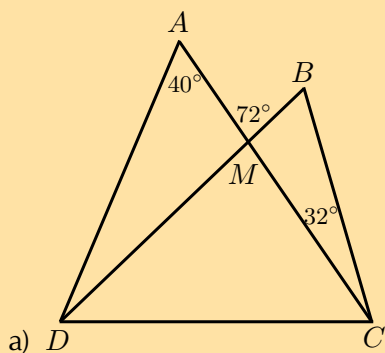
$$\begin{aligned}
 \hat{DBC} &= 90^\circ && (\angle \text{ in semi circle}) \\
 \text{and } \hat{E} &= 90^\circ && (\text{given}) \\
 \therefore \hat{DBC} &= \hat{E} \\
 \therefore ABDE &\text{ is a cyclic quadrilateral} && (\text{ext. } \angle \text{ equals int. opp. } \angle)
 \end{aligned}$$

Exercise 8 – 4: Cyclic quadrilaterals

1. Find the values of the unknown angles.



2. Prove that $ABCD$ is a cyclic quadrilateral:



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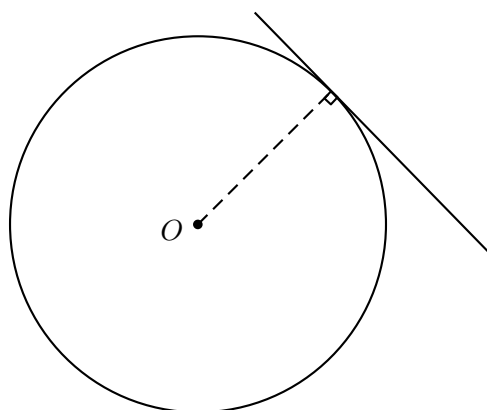
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Tangent line to a circle

A tangent is a line that touches the circumference of a circle at only one place. The radius of a circle is perpendicular to the tangent at the point of contact.



Theorem: Two tangents drawn from the same point outside a circle

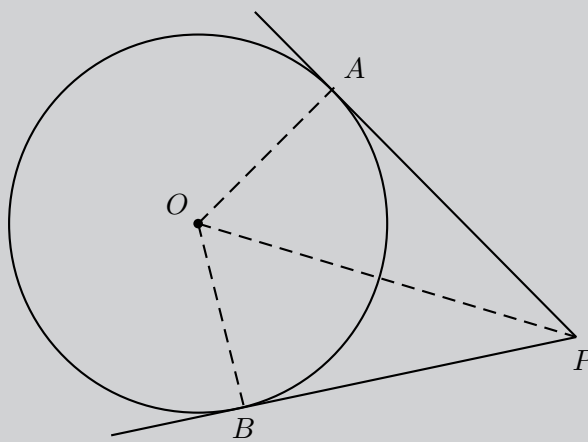
STATEMENT

If two tangents are drawn from the same point outside a circle, then they are equal in length.

(Reason: tangents from same point equal)

Given:

Circle with centre O and tangents PA and PB , where A and B are the respective points of contact for the two lines.



Required to prove:

$$AP = BP$$

PROOF

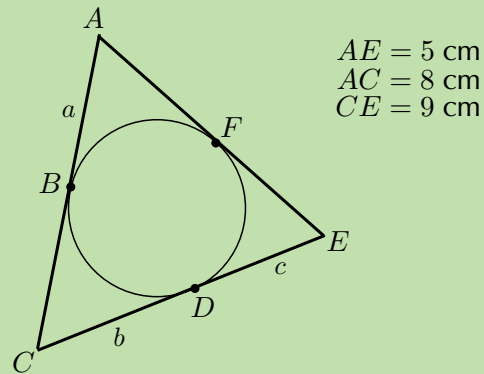
In $\triangle AOP$ and $\triangle BOP$,

$$\begin{aligned} \angle OAP &= \angle OBP = 90^\circ && \text{(tangent } \perp \text{ radius)} \\ AO &= BO && \text{(equal radii)} \\ OP &= OP && \text{(common side)} \\ \therefore \triangle AOP &\equiv \triangle BOP && \text{(RHS)} \\ \therefore AP &= BP \end{aligned}$$

Worked example 6: Tangents from the same point outside a circle

QUESTION

In the diagram below $AE = 5$ cm, $AC = 8$ cm and $CE = 9$ cm. Determine the values of a , b and c .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles on the diagram

Step 2: Solve for a , b and c

$$\begin{aligned} AB &= AF = a && \text{(tangents from } A) \\ EF &= ED = c && \text{(tangents from } E) \\ CB &= CD = b && \text{(tangents from } C) \\ \therefore AE &= a + c = 5 \\ \text{and } AC &= a + b = 8 \\ \text{and } CE &= b + c = 9 \end{aligned}$$

Step 3: Solve for the unknown variables using simultaneous equations

$$\begin{aligned} a + c &= 5 && \dots (1) \\ a + b &= 8 && \dots (2) \\ b + c &= 9 && \dots (3) \end{aligned}$$

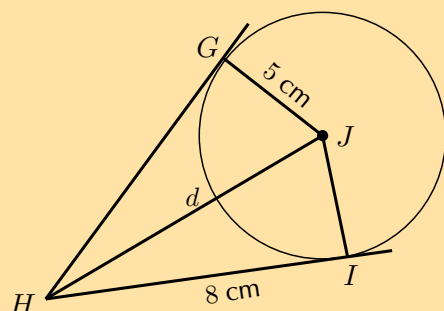
Subtract equation (1) from equation (2) and then substitute into equation (3):

$$\begin{aligned} (2) - (1) \quad b - c &= 8 - 5 \\ &= 3 \\ \therefore b &= c + 3 \\ \text{Substitute into (3)} \quad c + 3 + c &= 9 \\ 2c &= 6 \\ c &= 3 \\ \therefore a &= 2 \\ \text{and } b &= 6 \end{aligned}$$

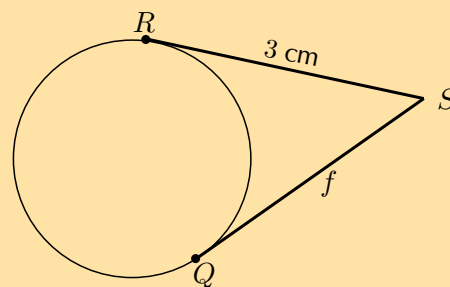
Exercise 8 – 5: Tangents to a circle

Find the values of the unknown lengths.

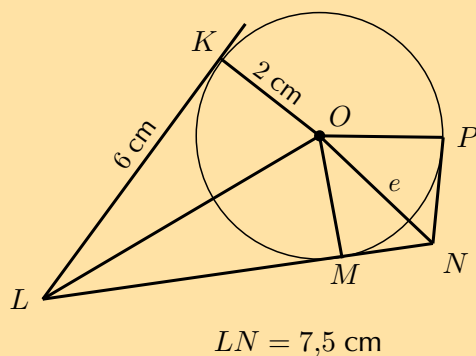
1.



3.



2.



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1. 236X 2. 236Y 3. 236Z



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Investigation: Tangent-chord theorem

Consider the diagrams given below:

Diagram 1

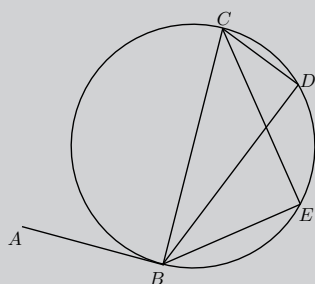


Diagram 2

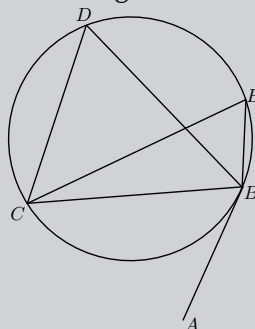
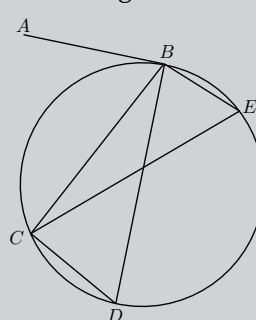


Diagram 3



1. Measure the following angles with a protractor and complete the table:

	Diagram 1	Diagram 2	Diagram 3
$\hat{A}BC =$			
$\hat{D} =$			
$\hat{E} =$			

2. Use your results to complete the following: the angle between a tangent to a circle and a chord is to the angle in the alternate segment.

Theorem: Tangent-chord theorem

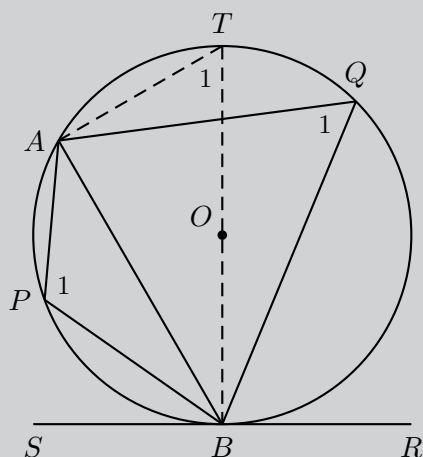
STATEMENT

The angle between a tangent to a circle and a chord drawn at the point of contact, is equal to the angle which the chord subtends in the alternate segment.

(Reason: tan. chord theorem)

Given:

Circle with centre O and tangent SR touching the circle at B . Chord AB subtends $\hat{P}_1$ and $\hat{Q}_1$.



Required to prove:

1. $\hat{ABR} = \hat{APB}$
2. $\hat{ABS} = \hat{AQB}$

PROOF

Draw diameter BT and join T to A .

Let $\hat{ATB} = T_1$.

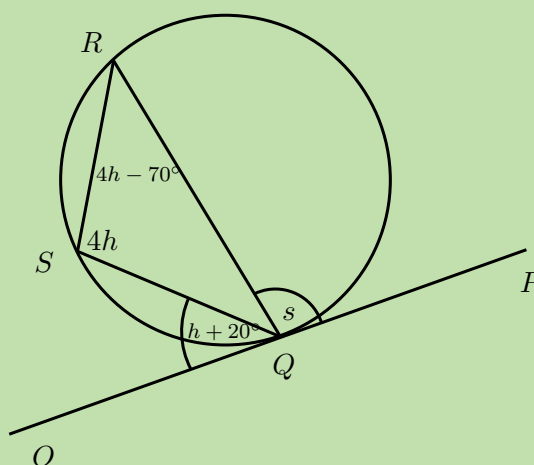
$$\begin{aligned} \hat{ABS} + \hat{ABT} &= 90^\circ && (\text{tangent} \perp \text{radius}) \\ \hat{BAT} &= 90^\circ && (\angle \text{ in semi circle}) \\ \therefore \hat{ABT} + T_1 &= 90^\circ && (\angle \text{ sum of } \triangle BAT) \\ \therefore \hat{ABS} &= T_1 \\ \text{but } \hat{Q_1} &= T_1 && (\angle \text{ s in same segment}) \\ \therefore \hat{Q_1} &= \hat{ABS} \end{aligned}$$

$$\begin{aligned} \hat{ABS} + \hat{ABR} &= 180^\circ && (\angle \text{ s on str. line}) \\ \hat{Q_1} + \hat{P_1} &= 180^\circ && (\text{opp. } \angle \text{ s cyclic quad. supp.}) \\ \therefore \hat{ABS} + \hat{ABR} &= \hat{Q_1} + \hat{P_1} \\ \text{and } \hat{ABS} &= \hat{Q_1} \\ \therefore \hat{ABR} &= \hat{P_1} \end{aligned}$$

Worked example 7: Tangent-chord theorem

QUESTION

Determine the values of h and s .



SOLUTION

Step 1: Use theorems and the given information to find all equal angles on the diagram

Step 2: Solve for h

$$\begin{aligned} \angle OSQ &= \angle SRQ && \text{(tangent chord theorem)} \\ h + 20^\circ &= 4h - 70^\circ \\ 90^\circ &= 3h \\ \therefore h &= 30^\circ \end{aligned}$$

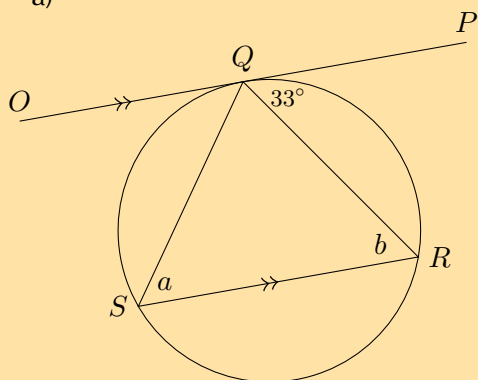
Step 3: Solve for s

$$\begin{aligned} \angle PQR &= \angle SQR && \text{(tangent chord theorem)} \\ s &= 4h \\ &= 4(30^\circ) \\ &= 120^\circ \end{aligned}$$

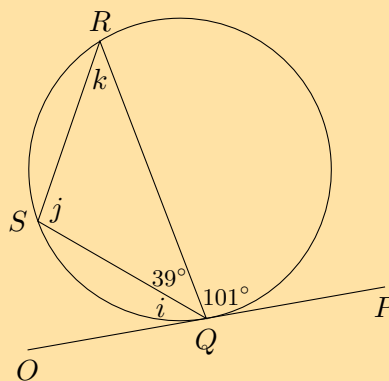
Exercise 8 – 6: Tangent-chord theorem

1. Find the values of the unknown letters, stating reasons.

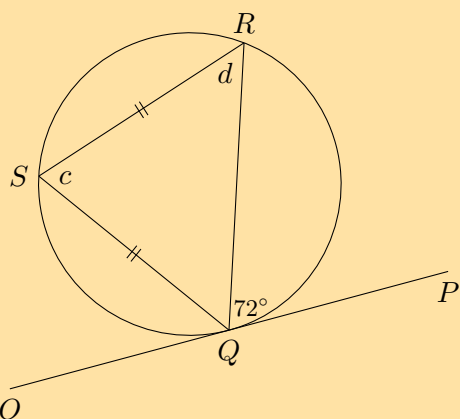
a)



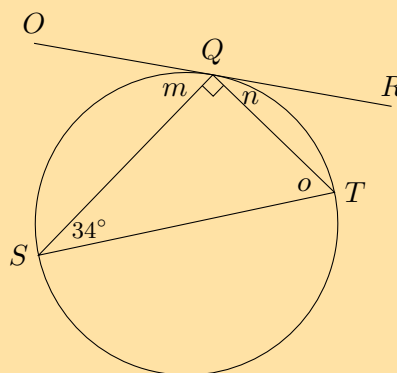
e)



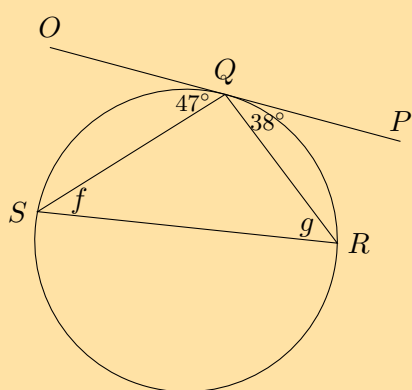
b)



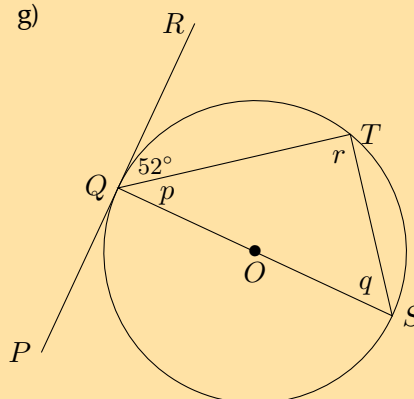
f)



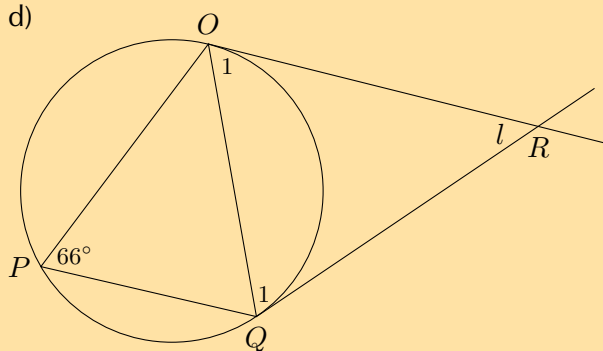
c)



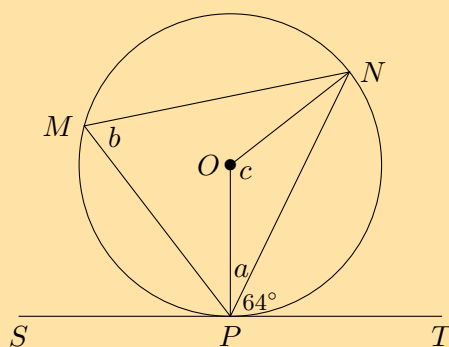
g)



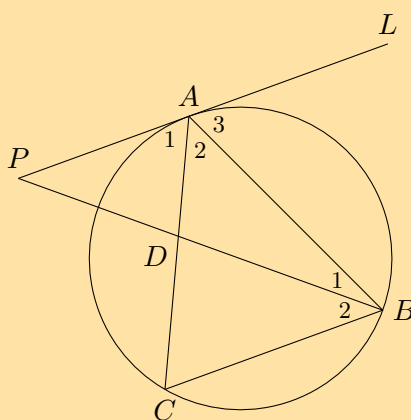
d)



2. O is the centre of the circle and SPT is a tangent, with $OP \perp ST$. Determine a , b and c , giving reasons.



3.



Given $AB = AC$, $AP \parallel BC$ and $\hat{A}_2 = \hat{B}_2$. Prove:

- PAL is a tangent to the circle ABC .
- AB is a tangent to the circle ADP .

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1a. [2372](#) 1b. [2373](#) 1c. [2374](#) 1d. [2375](#) 1e. [2376](#) 1f. [2377](#)
1g. [2378](#) 2. [2379](#) 3. [237B](#)



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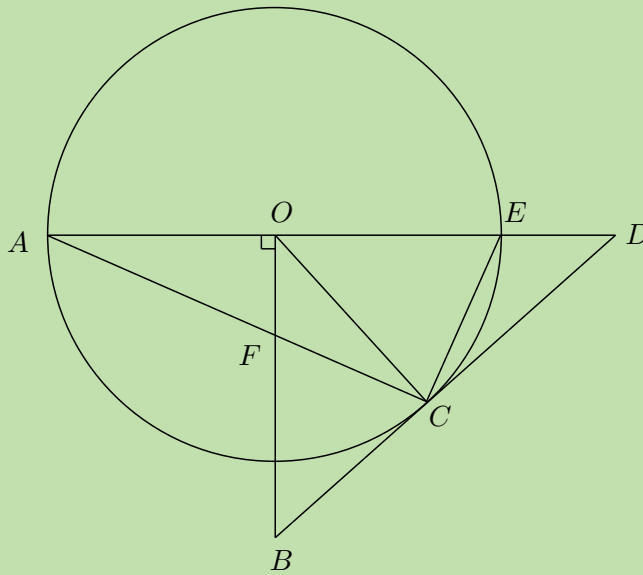
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Converse: tangent-chord theorem

If a line drawn through the end point of a chord forms an angle equal to the angle subtended by the chord in the alternate segment, then the line is a tangent to the circle.

(Reason: $\angle$ between line and chord $= \angle$ in alt. seg.)

QUESTION



BD is a tangent to the circle with centre O , with $BO \perp AD$.

Prove that:

1. $CFOE$ is a cyclic quadrilateral
2. $FB = BC$
3. $\angle A\hat{O}C = 2B\hat{F}C$
4. Will DC be a tangent to the circle passing through C, F, O and E ? Motivate your answer.

SOLUTION

Step 1: Prove $CFOE$ is a cyclic quadrilateral by showing opposite angles are supplementary

$$\begin{array}{ll}
 BO \perp OD & \text{(given)} \\
 \therefore \angle F\hat{O}E = 90^\circ & \\
 \angle F\hat{C}E = 90^\circ & \text{(\angle in semi circle)} \\
 \therefore CF\hat{O}E \text{ is a cyclic quad.} & \text{(opp. \angle s suppl.)}
 \end{array}$$

Step 2: Prove BFC is an isosceles triangle

To show that $FB = BC$ we first prove $\triangle BFC$ is an isosceles triangle by showing that $B\hat{F}C = B\hat{C}F$.

$$\begin{array}{ll}
 B\hat{C}F = C\hat{E}O & \text{(tangent-chord)} \\
 C\hat{E}O = B\hat{F}C & \text{(ext. \angle cyclic quad. } CF\hat{O}E) \\
 \therefore B\hat{F}C = B\hat{C}F & \\
 \therefore FB = BC & \text{(\triangle BFC isosceles)}
 \end{array}$$

Step 3: Prove $\hat{A}OC = 2\hat{B}FC$

$$\begin{aligned} A\hat{O}C &= 2A\hat{E}C && (\angle \text{ at centre } = 2\angle \text{ at circum.}) \\ \text{and } A\hat{E}C &= B\hat{F}C && (\text{ext. } \angle \text{ cyclic quad. } CFOE) \\ \therefore A\hat{O}C &= 2B\hat{F}C \end{aligned}$$

Step 4: Determine if DC is a tangent to the circle through C, F, O and E

Proof by contradiction.

Let us assume that DC is a tangent to the circle passing through the points C, F, O and E :

$$\therefore D\hat{C}E = C\hat{O}E \quad (\text{tangent-chord})$$

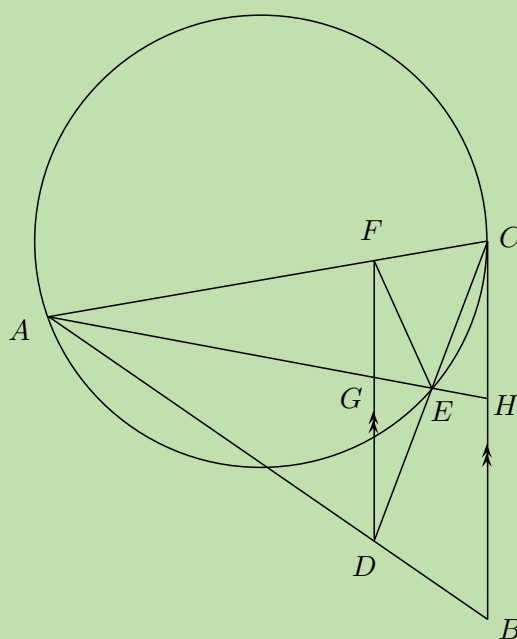
And using the circle with centre O and tangent BD we have that:

$$\begin{aligned} D\hat{C}E &= C\hat{A}E && \text{(tangent-chord)} \\ \text{but } C\hat{A}E &= \frac{1}{2}C\hat{O}E && (\angle \text{ at centre} = 2\angle \text{ at circum.}) \\ \therefore D\hat{C}E &\neq C\hat{O}E \end{aligned}$$

Therefore our assumption is not correct and we can conclude that DC is not a tangent to the circle passing through the points C, F, O and E .

Worked example 9: Applying the theorems

QUESTION



FD is drawn parallel to the tangent CB

Prove that:

1. $FADE$ is a cyclic quadrilateral
2. $\hat{FEA} = \hat{B}$

SOLUTION

Step 1: Prove $FADE$ is a cyclic quadrilateral using angles in the same segment

$$\begin{aligned} \hat{FDC} &= \hat{DCB} && (\text{alt. } \angle\text{s } FD \parallel CB) \\ \text{and } \hat{DCB} &= \hat{CAE} && (\text{tangent-chord}) \\ \therefore \hat{FDC} &= \hat{CAE} \\ \therefore FADE &\text{ is a cyclic quad.} && (\angle\text{s in same seg.}) \end{aligned}$$

Step 2: Prove $\hat{FEA} = \hat{B}$

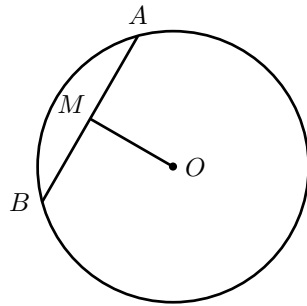
$$\begin{aligned} \hat{FDA} &= \hat{B} && (\text{corresp. } \angle\text{s } FD \parallel CB) \\ \text{and } \hat{FEA} &= \hat{FDA} && (\angle\text{s same seg. cyclic quad. } FADE) \\ \therefore \hat{FEA} &= \hat{B} \end{aligned}$$

8.3 Summary

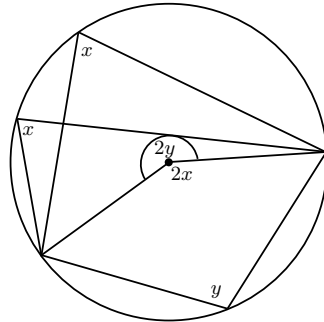
EMBJC

► See presentation: 237C at www.everythingmaths.co.za

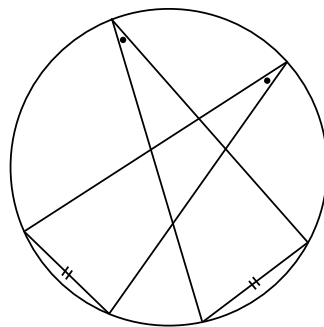
- **Arc** An arc is a portion of the circumference of a circle.
- **Chord** - a straight line joining the ends of an arc.
- **Circumference** - perimeter or boundary line of a circle.
- **Radius (r)** - any straight line from the centre of the circle to a point on the circumference.
- **Diameter** - a special chord that passes through the centre of the circle. A diameter is the length of a straight line segment from one point on the circumference to another point on the circumference, that passes through the centre of the circle.
- **Segment** A segment is a part of the circle that is cut off by a chord. A chord divides a circle into two segments.
- **Tangent** - a straight line that makes contact with a circle at only one point on the circumference.
- A tangent line is perpendicular to the radius, drawn at the point of contact with the circle.



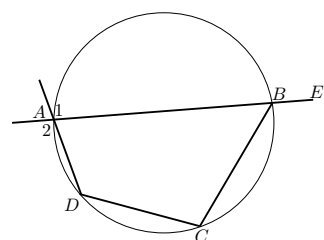
- If O is the centre and $OM \perp AB$, then $AM = MB$.
- If O is the centre and $AM = MB$, then $\hat{A}MO = \hat{B}MO = 90^\circ$.
- If $AM = MB$ and $OM \perp AB$, then $\Rightarrow MO$ passes through centre O .



If an arc subtends an angle at the centre of a circle and at the circumference, then the angle at the centre is twice the size of the angle at the circumference.

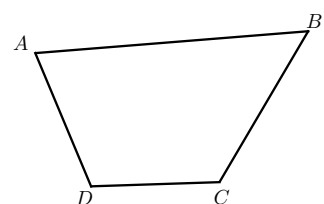


Angles at the circumference subtended by the same arc (or arcs of equal length) are equal.

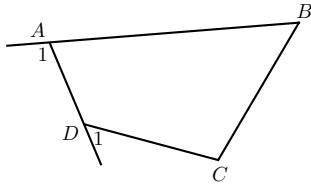


The four sides of a cyclic quadrilateral $ABCD$ are chords of the circle with centre O .

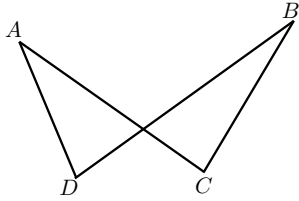
- $\hat{A} + \hat{C} = 180^\circ$ (opp. $\angle$ s supp.)
- $\hat{B} + \hat{D} = 180^\circ$ (opp. $\angle$ s supp.)
- $\hat{E}BC = \hat{D}$ (ext. $\angle$ cyclic quad.)
- $\hat{A}_1 = \hat{A}_2 = \hat{C}$ (vert. opp. $\angle$, ext. $\angle$ cyclic quad.)



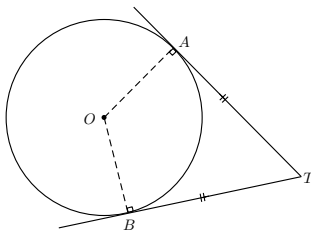
Proving a quadrilateral is cyclic: If $\hat{A} + \hat{C} = 180^\circ$ or $\hat{B} + \hat{D} = 180^\circ$, then $ABCD$ is a cyclic quadrilateral.



If $\hat{A}_1 = \hat{C}$ or $\hat{D}_1 = \hat{B}$, then $ABCD$ is a cyclic quadrilateral.

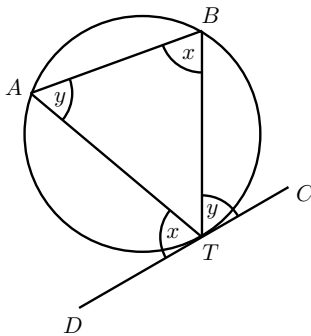


If $\hat{A} = \hat{B}$ or $\hat{C} = \hat{D}$, then $ABCD$ is a cyclic quadrilateral.



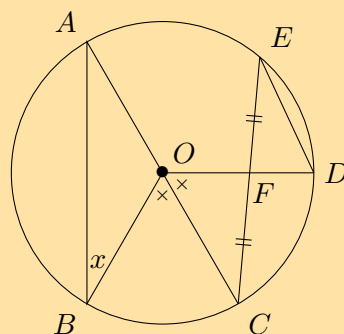
If AT and BT are tangents to circle O , then

- $OA \perp AT$ (tangent $\perp$ radius)
- $OB \perp BT$ (tangent $\perp$ radius)
- $TA = TB$ (tangents from same point equal)



- If DC is a tangent, then $\hat{DTA} = \hat{TBA}$ and $\hat{CTB} = \hat{TAB}$
- If $\hat{DTA} = \hat{TBA}$ or $\hat{CTB} = \hat{TAB}$, then DC is a tangent touching at T

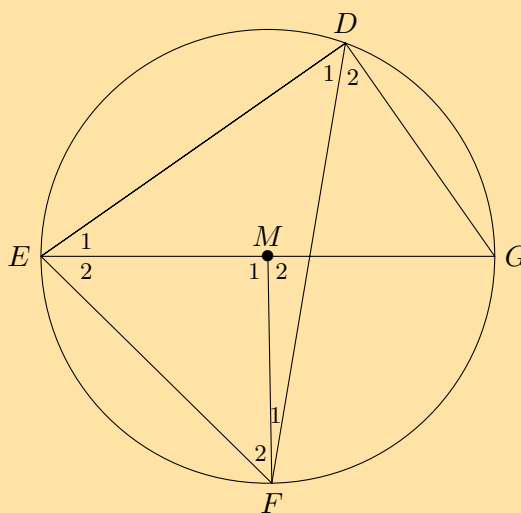
1.



AOC is a diameter of the circle with centre O . F is the mid-point of chord EC . $\hat{BOC} = \hat{COD}$ and $\hat{B} = x$. Express the following angles in terms of x , stating reasons:

- $\hat{A}$
- $\hat{COD}$
- $\hat{D}$

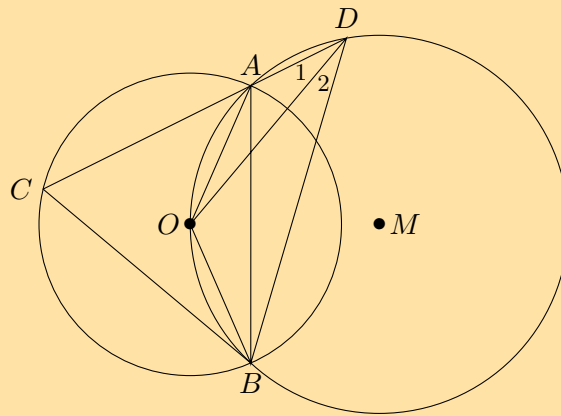
2.



D, E, F and G are points on circle with centre M . $\hat{F}_1 = 7^\circ$ and $\hat{D}_2 = 51^\circ$. Determine the sizes of the following angles, stating reasons:

- | | |
|----------------|----------------|
| a) $\hat{M}_1$ | d) $\hat{G}$ |
| b) $\hat{D}_1$ | e) $\hat{E}_1$ |
| c) $\hat{F}_2$ | |

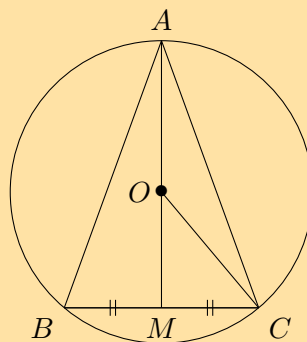
3.



O is a point on the circle with centre M . O is also the centre of a second circle. DA cuts the smaller circle at C and $\hat{D}_1 = x$. Express the following angles in terms of x , stating reasons:

- | | |
|----------------|----------------|
| a) $\hat{D}_2$ | d) $\hat{AOB}$ |
| b) $\hat{OAB}$ | |
| c) $\hat{OBA}$ | e) $\hat{C}$ |

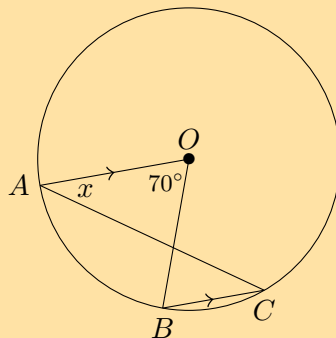
4.



O is the centre of the circle with radius 5 cm and chord $BC = 8$ cm. Calculate the lengths of:

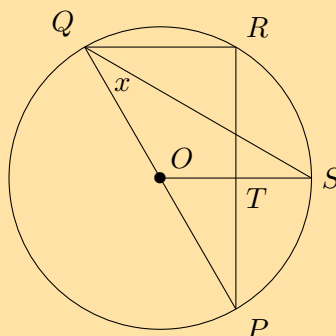
- OM
- AM
- AB

5.



$AO \parallel CB$ in circle with centre O . $\hat{AOB} = 70^\circ$ and $\hat{OAC} = x$. Calculate the value of x , giving reasons.

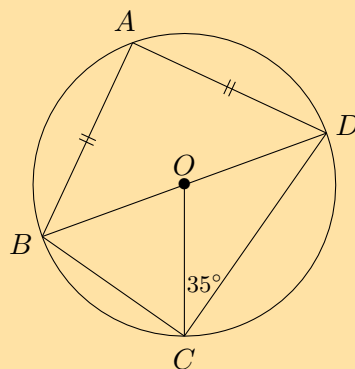
6.



PQ is a diameter of the circle with centre O . SQ bisects $\angle PQR$ and $\angle PQS = x$.

- Write down two other angles that are also equal to x .
- Calculate $\angle POS$ in terms of x , giving reasons.
- Prove that OS is a perpendicular bisector of PR .

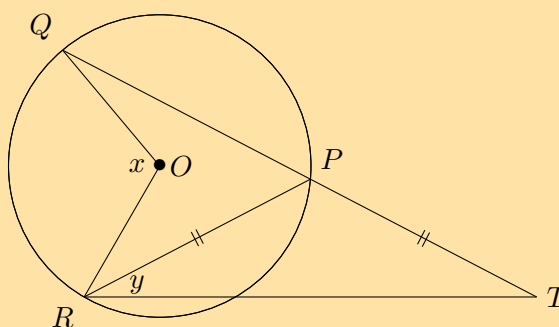
7.



BD is a diameter of the circle with centre O . $AB = AD$ and $\angle OCD = 35^\circ$. Calculate the value of the following angles, giving reasons:

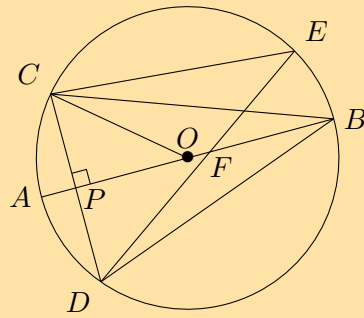
- | | |
|-----------------|-----------------|
| a) $\angle ODC$ | d) $\angle BAD$ |
| b) $\angle COD$ | e) $\angle ADB$ |
| c) $\angle CBD$ | |

8.



QP in the circle with centre O is protracted to T so that $PR = PT$. Express y in terms of x .

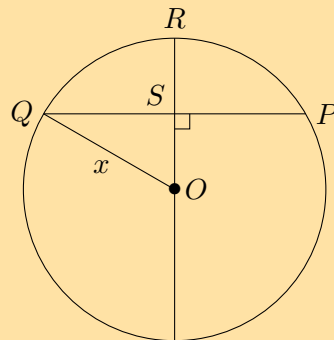
9.



O is the centre of the circle with diameter AB . $CD \perp AB$ at P and chord DE cuts AB at F . Prove that:

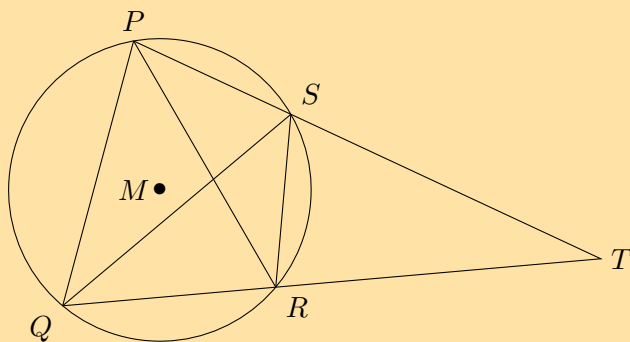
- $\angle CBP = \angle DPB$
- $\angle CED = 2\angle CBA$
- $\angle ABD = \frac{1}{2}\angle COA$

10.



In the circle with centre O , $OR \perp QP$, $PQ = 30$ mm and $RS = 9$ mm. Determine the length of OQ .

11.



P, Q, R and S are points on the circle with centre M . PS and QR are extended and meet at T . $PQ = PR$ and $\angle PQR = 70^\circ$.

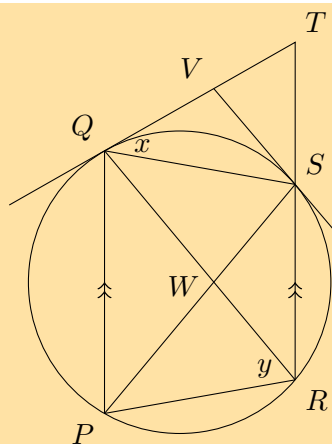
- Determine, stating reasons, three more angles equal to 70° .
- If $\angle QPS = 80^\circ$, calculate $\angle SRT$, $\angle STR$ and $\angle PQS$.
- Explain why PQ is a tangent to the circle QST at point Q .
- Determine $\angle PMQ$.

- $P\hat{C}Q = B\hat{A}P$
- $BAPC$ is a cyclic quadrilateral
- $AB = AC$

- a) $A\hat{B}T$
- b) $O\hat{B}A$
- c) $\hat{C}$

- Prove $AD = DC$
- Show that $\hat{ABC}$ is bisected by EB
- If $\hat{OEB} = x$, express $\hat{BAC}$ in terms of x
- Calculate the radius of the circle if $AC = 10$ cm and $DE = 1$ cm

15.

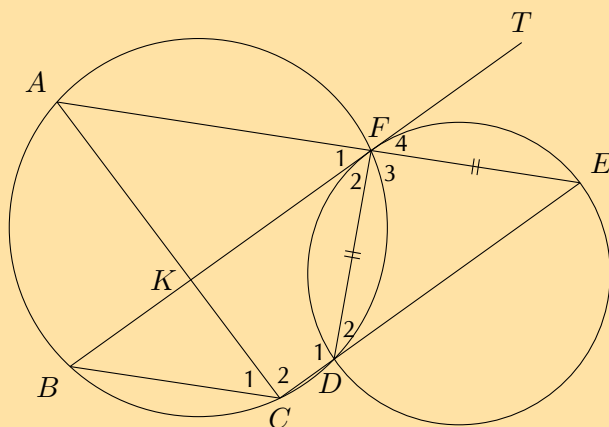


PQ and RS are chords of the circle and $PQ \parallel RS$. The tangent to the circle at Q meets RS protruded at T . The tangent at S meets QT at V . QS and PR are drawn.

Let $\hat{TQS} = x$ and $\hat{QRP} = y$. Prove that:

- $\hat{TVS} = 2\hat{QRS}$
- $QVSW$ is a cyclic quadrilateral
- $\hat{QPS} + \hat{T} = \hat{PRT}$
- W is the centre of the circle

16.



The two circles shown intersect at points F and D . BFT is a tangent to the smaller circle at F . Straight line AFE is drawn such that $DF = EF$. CDE is a straight line and chord AC and BF cut at K . Prove that:

- $BT \parallel CE$
- $BCEF$ is a parallelogram
- $AC = BF$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 237D | 2. 237F | 3. 237G | 4. 237H | 5. 237J | 6. 237K |
| 7. 237M | 8. 237N | 9. 237P | 10. 237Q | 11. 237R | 12. 237S |
| 13. 237T | 14. 237V | 15. 237W | 16. 237X | | |



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Finance, growth and decay

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9.1 Revision

EMBJD

Simple interest is the interest calculated only on the initial amount invested, the principal amount. Compound interest is the interest earned on the principal amount and on its accumulated interest. This means that interest is being earned on interest. The accumulated amount is the final amount; the sum of the principal amount and the amount of interest earned.

Formula for simple interest:

$$A = P(1 + in)$$

Formula for compound interest:

$$A = P(1 + i)^n$$

where

A = accumulated amount

P = principal amount

i = interest rate written as a decimal

n = time period in years

Worked example 1: Simple and compound interest

QUESTION

Sam wants to invest R 3450 for 5 years. Wise Bank offers a savings account which pays simple interest at a rate of 12,5% per annum, and Grand Bank offers a savings account paying compound interest at a rate of 10,4% per annum. Which bank account would give Sam the greatest accumulated balance at the end of the 5 year period?

SOLUTION

Step 1: Calculation using the simple interest formula

Write down the known variables and the simple interest formula

$$P = 3450$$

$$i = 0,125$$

$$n = 5$$

$$A = P(1 + in)$$

Substitute the values to determine the accumulated amount for the Wise Bank savings

account.

$$\begin{aligned}A &= 3450(1 + 0,125 \times 5) \\ &= \text{R } 5606,25\end{aligned}$$

Step 2: Calculation using the compound interest formula

Write down the known variables and the compound interest formula.

$$\begin{aligned}P &= 3450 \\ i &= 0,104 \\ n &= 5 \\ A &= P(1 + i)^n\end{aligned}$$

Substitute the values to determine the accumulated amount for the Grand Bank savings account.

$$\begin{aligned}A &= 3450(1 + 0,104)^5 \\ &= \text{R } 5658,02\end{aligned}$$

Step 3: Write the final answer

The Grand Bank savings account would give Sam the highest accumulated balance at the end of the 5 year period.

Worked example 2: Finding i

QUESTION

Bongani decides to put R 30 000 in an investment account. What compound interest rate must the investment account achieve for Bongani to double his money in 6 years? Give your answer correct to one decimal place.

SOLUTION

Step 1: Write down the known variables and the compound interest formula

$$\begin{aligned}A &= 60\,000 \\ P &= 30\,000 \\ n &= 6 \\ A &= P(1 + i)^n\end{aligned}$$

Step 2: Substitute the values and solve for i

$$\begin{aligned}60\,000 &= 30\,000(1+i)^6 \\ \frac{60\,000}{30\,000} &= (1+i)^6 \\ 2 &= (1+i)^6 \\ \sqrt[6]{2} &= 1+i \\ \sqrt[6]{2} - 1 &= i \\ \therefore i &= 0,122 \dots\end{aligned}$$

Step 3: Write the final answer and comment

We round up to a rate of 12,3% p.a. to make sure that Bongani doubles his investment.

Exercise 9 – 1: Revision

1. Determine the value of an investment of R 10 000 at 12,1% p.a. simple interest for 3 years.
2. Calculate the value of R 8000 invested at 8,6% p.a. compound interest for 4 years.
3. Calculate how much interest John will earn if he invests R 2000 for 4 years at:
 - a) 6,7% p.a. simple interest
 - b) 5,4% p.a. compound interest
4. The value of an investment grows from R 2200 to R 3850 in 8 years. Determine the simple interest rate at which it was invested.
5. James had R 12 000 and invested it for 5 years. If the value of his investment is R 15 600, what compound interest rate did it earn?

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1. 237Y 2. 237Z 3. 2382 4. 2383 5. 2384



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As soon as a new car leaves the dealership, its value decreases and it is considered “second-hand”. Vehicles, equipment, machinery and other similar assets, all lose value over time as a result of usage and age. This loss in value is called depreciation. Assets that have a relatively long useful lifetime, such as machines, trucks, farming equipment etc., depreciate slower than assets like office equipment, computers, furniture etc. which need to be replaced more often and therefore depreciate more quickly.

Depreciation is used to calculate the value of a company’s assets, which determines how much tax a company must pay. Companies can take depreciation into account as an expense, and thereby reduce their taxable income. A lower taxable income means that the company will pay less income tax to SARS (South African Revenue Service).

We can calculate two different kinds of depreciation: simple decay and compound decay. Decay is also a term used to describe a reduction or decline in value. Simple decay is also called straight-line depreciation and compound decay can also be referred to as reducing-balance depreciation. In the straight-line method the value of the asset is reduced by a constant amount each year, which is calculated on the principal amount. In reducing-balance depreciation we calculate the depreciation on the reduced value of the asset. This means that the value of an asset decreases by a different amount each year.

Investigation: Simple and compound depreciation

1. Mr. Sontange buys an Opel Fiesta for R 72 000. He expects that the value of the car will depreciate by R 6000 every year. He draws up a table to calculate the depreciated value of his Opel Fiesta.

Complete Mr. Sontange’s table of values for the 7 year period:

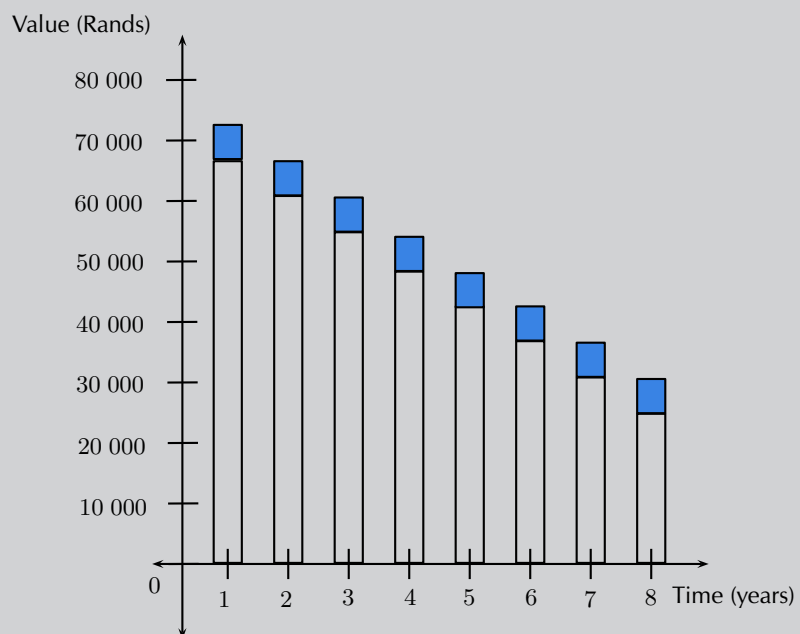
Year	Value at beginning of year	Depreciation amount	Value at end of year
1	R 72 000	R 6000	R 66 000
2	R 66 000	R 6000	
3			
4			
5			
6			
7			

2. His son, David, does not agree that the value of the car will reduce by the same amount each year. David thinks that the car will depreciate by 10% every year.

Complete David’s table of values:

Year	Value at beginning of year	Depreciation amount	Value at end of year
1	R 72 000	R 7200	R 64 800
2	R 64 800	R 6480	
3			
4			
5			
6			
7			

3. Compare and discuss the results of the two different tables.
4. Consider the graph below, which represents Mr. Sontange's table of values:



- a) Draw a similar graph using David's table of values.
- b) Interpret the two graphs and discuss the differences between them.
- c) Explain how the graphs can be used to determine the total depreciation in each case.
- d)
 - i. Draw two new graphs by plotting the maximum value of each bar.
 - ii. Join the points with a line to show the general trend.
 - iii. Is it mathematically correct to join these points? Explain your answer.

Worked example 3: Straight-line depreciation**QUESTION**

A new smartphone costs R 6000 and depreciates at 22% p.a. on a straight-line basis. Determine the value of the smartphone at the end of each year over a 4 year period.

SOLUTION**Step 1: Calculate depreciation amount**

$$\begin{aligned}\text{Depreciation} &= 6000 \times \frac{22}{100} \\ &= 1320\end{aligned}$$

Therefore the smartphone depreciates by R 1320 every year.

Step 2: Complete a table of values

Year	Value at beginning of year	Depreciation amount	Value at end of year
1	R 6000	R 1320	R 4680
2	R 4680	R 1320	R 3360
3	R 3360	R 1320	R 2040
4	R 2040	R 1320	R 720

We notice that

$$\text{Total depreciation} = P \times i \times n$$

where

P = principal amount

i = interest rate written as a decimal

n = time period in years

Therefore the depreciated value of the asset (also called the book value) can be calculated as:

$$A = P(1 - in)$$

Note the similarity to the simple interest formula $A = P(1 + in)$. Interest increases the value of the principal amount, whereas with simple decay, depreciation reduces the value of the principal amount.

Important: to get an accurate answer do all calculations in one step on your calculator. Do not round off answers in your calculations until the final answer. In the worked examples in this chapter, we use dots to show that the answer has not been rounded off. We always round the final answer to two decimal places (cents).

Worked example 4: Straight-line depreciation method

QUESTION

A car is valued at R 240 000. If it depreciates at 15% p.a. using straight-line depreciation, calculate the value of the car after 5 years.

SOLUTION

Step 1: Write down the known variables and the simple decay formula

$$P = 240\,000$$

$$i = 0,15$$

$$n = 5$$

$$A = P(1 - in)$$

Step 2: Substitute the values and solve for A

$$\begin{aligned} A &= 240\,000(1 - 0,15 \times 5) \\ &= 240\,000(0,25) \\ &= 60\,000 \end{aligned}$$

Step 3: Write the final answer

At the end of 5 years, the car is worth R 60 000.

Worked example 5: Simple decay

QUESTION

A small business buys a photocopier for R 12 000. For the tax return the owner depreciates this asset over 3 years using a straight-line depreciation method. What amount will he fill in on his tax form at the end of each year?

SOLUTION

Step 1: Write down the known variables

The owner of the business wants the photocopier to have a book value of R 0 after 3 years.

$$A = 0$$

$$P = 12\,000$$

$$n = 3$$

Therefore we can calculate the annual depreciation as

$$\begin{aligned}\text{Depreciation} &= \frac{P}{n} \\ &= \frac{12\,000}{3} \\ &= \text{R } 4000\end{aligned}$$

Step 2: Determine the book value at the end of each year

$$\begin{aligned}\text{Book value end of first year} &= 12\,000 - 4000 \\ &= \text{R } 8000\end{aligned}$$

$$\begin{aligned}\text{Book value end of second year} &= 8000 - 4000 \\ &= \text{R } 4000\end{aligned}$$

$$\begin{aligned}\text{Book value end of third year} &= 4000 - 4000 \\ &= \text{R } 0\end{aligned}$$

Exercise 9 – 2: Simple decay

1. A business buys a truck for R 560 000. Over a period of 10 years the value of the truck depreciates to R 0 using the straight-line method. What is the value of the truck after 8 years?
2. Harry wants to buy his grandpa's donkey for R 800. His grandpa is quite pleased with the offer, seeing that it only depreciated at a rate of 3% per year using the straight-line method. Grandpa bought the donkey 5 years ago. What did grandpa pay for the donkey then?
3. Seven years ago, Rocco's drum kit cost him R 12 500. It has now been valued at R 2300. What rate of simple depreciation does this represent?
4. Fiona buys a DStv satellite dish for R 3000. Due to weathering, its value depreciates simply at 15% per annum. After how long will the satellite dish have a book value of zero?

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Worked example 6: Reducing-balance depreciation**QUESTION**

A second-hand farm tractor worth R 60 000 has a limited useful life of 5 years and depreciates at 20% p.a. on a reducing-balance basis. Determine the value of the tractor at the end of each year over the 5 year period.

SOLUTION**Step 1: Write down the known variables**

$$P = 60\,000$$

$$i = 0,2$$

$$n = 5$$

When we calculate depreciation using the reducing-balance method:

1. the depreciation amount changes for each year.
2. the depreciation amount gets smaller each year.
3. the book value at the end of a year becomes the principal amount for the next year.
4. the asset will always have some value (the book value will never equal zero).

Step 2: Complete a table of values

Year	Book value	Depreciation	Value at end of year
1	R 60 000	$60\,000 \times 0,2 = 12\,000$	R 48 000
2	R 48 000	$48\,000 \times 0,2 = 9600$	R 38 400
3	R 38 400	$38\,400 \times 0,2 = 7680$	R 30 720
4	R 30 720	$30\,720 \times 0,2 = 6144$	R 24 576
5	R 24 576	$24\,576 \times 0,2 = 4915,20$	R 19 660,80

Notice in the example above that we could also write the book value at the end of each year as:

$$\text{Book value end of first year} = 60\,000(1 - 0,2)$$

$$\text{Book value end of second year} = 48\,000(1 - 0,2) = 60\,000(1 - 0,2)^2$$

$$\text{Book value end of third year} = 38\,400(1 - 0,2) = 60\,000(1 - 0,2)^3$$

$$\text{Book value end of fourth year} = 30\,720(1 - 0,2) = 60\,000(1 - 0,2)^4$$

$$\text{Book value end of fifth year} = 24\,576(1 - 0,2) = 60\,000(1 - 0,2)^5$$

Using the formula for simple decay and the observed pattern in the calculation above, we obtain the following formula for compound decay:

$$A = P(1 - i)^n$$

where

A = book value or depreciated value

P = principal amount

i = interest rate written as a decimal

n = time period in years

Again, notice the similarity to the compound interest formula $A = P(1 + i)^n$.

Worked example 7: Reducing-balance depreciation

QUESTION

The number of pelicans at the Berg river mouth is decreasing at a compound rate of 12% p.a. If there are currently 3200 pelicans in the wetlands of the Berg river mouth, what will the population be in 5 years?

SOLUTION

Step 1: Write down the known variables and the compound decay formula

$$P = 3200$$

$$i = 0,12$$

$$n = 5$$

$$A = P(1 - i)^n$$

Step 2: Substitute the values and solve for A

$$A = 3200(1 - 0,12)^5$$

$$= 3200(0,88)^5$$

$$= 1688,7421 \dots$$

Step 3: Write the final answer

In 5 years, the pelican population will be approximately 1689.

QUESTION

1. A school buys a minibus for R 950 000, which depreciates at 13,5% per annum. Determine the value of the minibus after 3 years if the depreciation is calculated:
 - a) on a straight-line basis.
 - b) on a reducing-balance basis.
2. Which is the better option?

SOLUTION

Step 1: Write down known variables

$$P = 950\,000$$

$$i = 0,135$$

$$n = 3$$

Step 2: Use the simple decay formula and solve for A

$$\begin{aligned}A &= 950\,000(1 - 3 \times 0,135) \\&= 950\,000(0,865) \\&= 565\,250 \\ \therefore A &= \text{R } 565\,250\end{aligned}$$

Step 3: Use the compound decay formula and solve for A

$$\begin{aligned}A &= 950\,000(1 - 0,135)^3 \\&= 950\,000(0,865)^3 \\&= 614\,853,89 \\ \therefore A &= \text{R } 614\,853,89\end{aligned}$$

Step 4: Interpret the answers

After a period of 3 years, the value of the minibus calculated on the straight-line method is less than the value of the minibus calculated on the reducing-balance method. The value of the minibus depreciated less on the reducing-balance basis because the amount of depreciation is calculated on a smaller amount every year, whereas the straight-line method is based on the full value of the minibus every year.

Worked example 9: Compound depreciation

QUESTION

Farmer Jack bought a tractor and it has depreciated by 20% p.a. on a reducing-balance basis. If the current value of the tractor is R 52 429, calculate how much Farmer Jack paid for his tractor if he bought it 7 years ago.

SOLUTION

Step 1: Write down known variables and compound decay formula

$$A = 52\,429$$

$$i = 0,2$$

$$n = 7$$

$$A = P(1 - i)^n$$

Step 2: Substitute the values and solve for P

$$\begin{aligned} 52\,429 &= P(1 - 0,2)^7 \\ &= P(0,8)^7 \\ \therefore P &= \frac{52\,429}{(0,8)^7} \\ &= 250\,000,95 \dots \end{aligned}$$

Step 3: Write the final answer

7 years ago, Farmer Jack paid R 250 000 for his tractor.

Exercise 9 – 3: Compound depreciation

1. Jwayelani buys a truck for R 89 000 and depreciates it by 9% p.a. using the compound depreciation method. What is the value of the truck after 14 years?
2. The number of cormorants at the Amanzimtoti river mouth is decreasing at a compound rate of 8% p.a. If there are now 10 000 cormorants, how many will there be in 18 years' time?
3. On January 1, 2008 the value of my Kia Sorento is R 320 000. Each year after that, the car's value will decrease 20% of the previous year's value. What is the value of the car on January 1, 2012?

4. The population of Bonduel decreases at a reducing-balance rate of 9,5% per annum as people migrate to the cities. Calculate the decrease in population over a period of 5 years if the initial population was 2 178 000.
5. A 20 kg watermelon consists of 98% water. If it is left outside in the sun it loses 3% of its water each day. How much does it weigh after a month of 31 days?
6. Richard bought a car 15 years ago and it depreciated by 17% p.a. on a compound depreciation basis. How much did he pay for the car if it is now worth R 5256?

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Finding i

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Worked example 10: Finding i for simple decay

QUESTION

After 4 years, the value of a computer is halved. Assuming simple decay, at what annual rate did it depreciate? Give your answer correct to two decimal places.

SOLUTION

Step 1: Write down known variables and simple decay formula

Let the value of the computer be x , therefore:

$$A = \frac{x}{2}$$

$$P = x$$

$$n = 4$$

$$A = P(1 - in)$$

Step 2: Substitute the values and solve for i

$$\begin{aligned}\frac{x}{2} &= x(1 - 3i) \\ \frac{1}{2} &= 1 - 3i \\ \therefore 3i &= 1 - \frac{1}{2} \\ \therefore i &= 0,1667\end{aligned}$$

Step 3: Write the final answer

The computer depreciated at a rate of 16,67% p.a.

Worked example 11: Finding i for compound decay

QUESTION

Cristina bought a fridge at the beginning of 2009 for R 8999 and sold it at the end of 2011 for R 4500. At what rate did the value of her fridge depreciate assuming a reducing-balance method? Give your answer correct to two decimal places.

SOLUTION

Step 1: Write down known variables and compound decay formula

$$\begin{aligned}A &= 4500 \\ P &= 8999 \\ n &= 3 \\ A &= P(1 - i)^n\end{aligned}$$

Step 2: Substitute the values and solve for i

$$\begin{aligned}4500 &= 8999(1 - i)^3 \\ \frac{4500}{8999} &= (1 - i)^3 \\ \sqrt[3]{\frac{4500}{8999}} &= 1 - i \\ \therefore i &= 1 - \sqrt[3]{\frac{4500}{8999}} \\ &= 0,206\end{aligned}$$

Step 3: Write the final answer

Cristina's fridge depreciated at a rate of 20,6% p.a.

Exercise 9 – 4: Finding i

1. A machine costs R 45 000 and has a scrap value of R 9000 after 10 years. Determine the annual rate of depreciation if it is calculated on the reducing balance method.
2. After 15 years, an aeroplane is worth $\frac{1}{6}$ of its original value. At what annual rate was depreciation compounded?
3. Mr. Mabula buys furniture for R 20 000. After 6 years he sells the furniture for R 9300. Calculate the annual compound rate of depreciation of the furniture.
4. Ayanda bought a new car 7 years ago for double what it is worth today. At what yearly compound rate did her car depreciate?

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9.3 Timelines

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Interest can be compounded more than once a year. For example, an investment can be compounded monthly or quarterly. Below is a table of compounding terms and their corresponding numeric value (p). When amounts are compounded more than once per annum, we multiply the number of years by p and we also divide the interest rate by p .

Term	p
yearly / annually	1
half-yearly / bi-annually	2
quarterly	4
monthly	12
weekly	52
daily	365

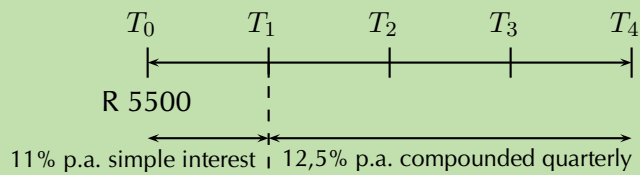
Worked example 12: Timelines

QUESTION

R 5500 is invested for a period of 4 years in a savings account. For the first year, the investment grows at a simple interest rate of 11% p.a. and then at a rate of 12,5% p.a. compounded quarterly for the rest of the period. Determine the value of the investment at the end of the 4 years.

SOLUTION

Step 1: Draw a timeline and write down known variables



In the timeline above, the intervals are given in years. For example, T_0 is the start of the investment, T_1 is the end of the first year and T_4 is the end of the fourth year.

Step 2: Use the simple interest formula to calculate A at T_1

$$\begin{aligned}
 A &= P(1 + in) \\
 &= 5500(1 + 0,11) \\
 &= \text{R } 6105
 \end{aligned}$$

Step 3: Use the compound interest formula to calculate A at T_4

The investment is compounded quarterly, therefore:

$$\begin{aligned}
 n &= 3 \times 4 \\
 &= 12 \\
 \text{and } i &= \frac{0,125}{4}
 \end{aligned}$$

Also notice that the accumulated amount at the end of the first year becomes the principal amount at the beginning of the second year.

$$\begin{aligned}
 A &= P(1 + i)^n \\
 &= 6105 \left(1 + \frac{0,125}{4}\right)^{12} \\
 &= \text{R } 8831,88
 \end{aligned}$$

Step 4: Write the final answer

The value of the investment at the end of the 4 years is R 8831,88.

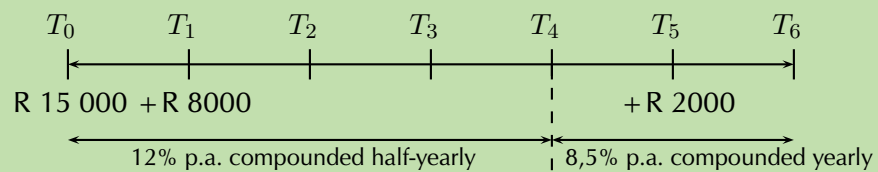
Worked example 13: Timelines

QUESTION

R 150 000 is deposited in an investment account for a period of 6 years at an interest rate of 12% p.a. compounded half-yearly for the first 4 years and then 8,5% p.a. compounded yearly for the rest of the period. A deposit of R 8000 is made into the account after the first year and then another deposit of R 2000 is made 5 years after the initial investment. Calculate the value of the investment at the end of the 6 years.

SOLUTION

Step 1: Draw a timeline and write down known variables



Remember to show when the additional deposits of R 8000 and R 2000 were made into the account. It is very important to note that the interest rate changes at T_4 .

We break this question down into parts and consider each amount separately.

Step 2: The initial deposit at T_0

Between T_0 and T_4 :

We notice that interest for the first 4 years is compounded half-yearly, therefore:

$$\begin{aligned}n_1 &= 4 \times 2 \\&= 8 \\ \text{and } i_1 &= \frac{0,12}{2}\end{aligned}$$

Between T_4 and T_6 :

$$\begin{aligned}n_2 &= 2 \\ \text{and } i_2 &= 0,085\end{aligned}$$

Therefore the total growth of the initial deposit over the 6 years is:

$$\begin{aligned}A &= P(1 + i_1)^{n_1}(1 + i_2)^{n_2} \\&= 150\,000 \left(1 + \frac{0,12}{2}\right)^8 (1 + 0,085)^2\end{aligned}$$

Step 3: The deposit at T_1

Between T_1 and T_4 :

Interest on this deposit is compounded half-yearly for 3 years, therefore:

$$\begin{aligned}n_3 &= 3 \times 2 \\&= 6 \\ \text{and } i_3 &= \frac{0,12}{2}\end{aligned}$$

Between T_4 and T_6 :

$$\begin{aligned}n_4 &= 2 \\ \text{and } i_4 &= 0,085\end{aligned}$$

Therefore the total growth of the deposit over the 5 years is:

$$\begin{aligned}A &= P(1 + i_3)^{n_3}(1 + i_4)^{n_4} \\&= 8000 \left(1 + \frac{0,12}{2}\right)^6 (1 + 0,085)^2\end{aligned}$$

Step 4: The deposit at T_5

Accumulate interest for only 1 year:

$$\begin{aligned}A &= P(1 + i)^n \\&= 2000(1 + 0,085)^1\end{aligned}$$

Step 5: Determine the total calculation

To get as accurate an answer as possible, we do the the calculation on the calculator in one step. Using the memory and answer recall function on the calculator, we avoid rounding off until we get the final answer.

$$\begin{aligned}A &= 150\,000 \left(1 + \frac{0,12}{2}\right)^8 (1 + 0,085)^2 \\&\quad + 8000 \left(1 + \frac{0,12}{2}\right)^6 (1 + 0,085)^2 + 2000(1 + 0,085)^1 \\&= \text{R } 296\,977,00\end{aligned}$$

Step 6: Write the final answer

The value of the investment at the end of the 6 years is R 296 977,00.

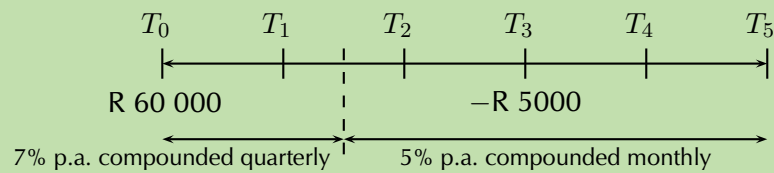
Worked example 14: Timelines

QUESTION

R 60 000 is invested in an account which offers interest at 7% p.a. compounded quarterly for the first 18 months. Thereafter the interest rate changes to 5% p.a. compounded monthly. Three years after the initial investment, R 5000 is withdrawn from the account. How much will be in the account at the end of 5 years?

SOLUTION

Step 1: Draw a timeline and write down known variables



Remember to show when the withdrawal of R 5000 was taken out of the account. It is also important to note that the interest rate changes after 18 months ($T_{1\frac{1}{2}}$).

We break this question down into parts and consider each amount separately.

Step 2: The initial deposit at T_0

Interest for the first 1,5 years is compounded quarterly, therefore:

$$\begin{aligned}n_1 &= 1,5 \times 4 \\&= 6 \\ \text{and } i_1 &= \frac{0,07}{4}\end{aligned}$$

Interest for the remaining 3,5 years is compounded monthly, therefore:

$$\begin{aligned}n_2 &= 3,5 \times 12 \\&= 42 \\ \text{and } i_2 &= \frac{0,05}{12}\end{aligned}$$

Therefore the total growth of the initial deposit over the 5 years is:

$$\begin{aligned}A &= P(1 + i_1)^{n_1}(1 + i_2)^{n_2} \\&= 60\,000 \left(1 + \frac{0,07}{4}\right)^6 \left(1 + \frac{0,05}{12}\right)^{42}\end{aligned}$$

Step 3: The withdrawal at T_3

We calculate the interest that the R 5000 would have earned if it had remained in the account:

$$\begin{aligned}n &= 2 \times 12 \\&= 24 \\ \text{and } i &= \frac{0,05}{12}\end{aligned}$$

Therefore we have that:

$$\begin{aligned}A &= P(1 + i)^n \\&= 5000 \left(1 + \frac{0,05}{12}\right)^{24}\end{aligned}$$

Step 4: Determine the total calculation

We subtract the withdrawal and the interest it would have earned from the accumulated amount at the end of the 5 years:

$$\begin{aligned}A &= 60\,000 \left(1 + \frac{0,07}{4}\right)^6 \left(1 + \frac{0,05}{12}\right)^{42} - 5000 \left(1 + \frac{0,05}{12}\right)^{24} \\&= \text{R } 73\,762,19\end{aligned}$$

Step 5: Write the final answer

The value of the investment at the end of the 5 years is R 73 762,19.

Exercise 9 – 5: Timelines

1. After a 20-year period Josh's lump sum investment matures to an amount of R 313 550. How much did he invest if his money earned interest at a rate of 13,65% p.a. compounded half yearly for the first 10 years, 8,4% p.a. compounded quarterly for the next five years and 7,2% p.a. compounded monthly for the remaining period?
2. Sindisiwe wants to buy a motorcycle. The cost of the motorcycle is R 55 000. In 1998 Sindisiwe opened an account at Sutherland Bank with R 16 000. Then in 2003 she added R 2000 more into the account. In 2007 Sindisiwe made another change: she took R 3500 from the account. If the account pays 6% p.a. compounded half-yearly, will Sindisiwe have enough money in the account at the end of 2012 to buy the motorcycle?
3. A loan has to be returned in two equal semi-annual instalments. If the rate of interest is 16% per annum, compounded semi-annually and each instalment is R 1458, find the sum borrowed.

4. A man named Phillip invests R 10 000 into an account at North Bank at an interest rate of 7,5% p.a. compounded monthly. After 5 years the bank changes the interest rate to 8% p.a. compounded quarterly. How much money will Phillip have in his account 9 years after the original deposit?
5. R 75 000 is invested in an account which offers interest at 11% p.a. compounded monthly for the first 24 months. Then the interest rate changes to 7,7% p.a. compounded half-yearly. If R 9000 is withdrawn from the account after one year and then a deposit of R 3000 is made three years after the initial investment, how much will be in the account at the end of 6 years?
6. Christopher wants to buy a computer, but right now he doesn't have enough money. A friend told Christopher that in 5 years the computer will cost R 9150. He decides to start saving money today at Durban United Bank. Christopher deposits R 5000 into a savings account with an interest rate of 7,95% p.a. compounded monthly. Then after 18 months the bank changes the interest rate to 6,95% p.a. compounded weekly. After another 6 months, the interest rate changes again to 7,92% p.a. compounded two times per year. How much money will Christopher have in the account after 5 years, and will he then have enough money to buy the computer?

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9.4 Nominal and effective interest rates

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We have seen that although interest is quoted as a percentage per annum it can be compounded more than once a year. We therefore need a way of comparing interest rates. For example, is an annual interest rate of 8% compounded quarterly higher or lower than an interest rate of 8% p.a. compounded yearly?

Investigation: Nominal and effective interest rates

1. Calculate the accumulated amount at the end of one year if R 1000 is invested at 8% p.a. compound interest:

$$A = P(1 + i)^n$$

$$= \dots\dots$$

2. Calculate the value of R 1000 if it is invested for one year at 8% p.a. compounded:

Frequency	Calculation	Accumulated amount	Interest amount
half-yearly	$A = 1000 \left(1 + \frac{0,08}{2}\right)^{1 \times 2}$	R 1081,60	R 81,60
quarterly			
monthly			
weekly			
daily			

3. Use your results from the table above to calculate the effective rate that the investment of R 1000 earns in one year:

Frequency	Accumulated amount	Calculation	Effective interest rate
half-yearly	R 1081,60	$1081,60 = 1000(1 + i)$ $\frac{1081,60}{1000} = 1 + i$ $\frac{1081,60}{1000} - 1 = i$ $\therefore i = 0,0816$	$i = 8,16\%$
quarterly			
monthly			
weekly			
daily			

4. If you wanted to borrow R 10 000 from the bank, would it be better to pay it back at an interest rate of 22% p.a. compounded quarterly or 22% compounded monthly? Show your calculations.

An interest rate compounded more than once a year is called the nominal interest rate. In the investigation above, we determined that the nominal interest rate of 8% p.a. compounded half-yearly is actually an effective rate of 8,16% p.a.

Given a nominal interest rate $i^{(m)}$ compounded at a frequency of m times per year and the effective interest rate i , the accumulated amount calculated using both interest rates will be equal so we can write:

$$P(1 + i) = P \left(1 + \frac{i^{(m)}}{m}\right)^m$$

$$\therefore 1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

Worked example 15: Nominal and effective interest rates

QUESTION

Interest on a credit card is quoted as 23% p.a. compounded monthly. What is the effective annual interest rate? Give your answer correct to two decimal places.

SOLUTION

Step 1: Write down the known variables

Interest is being added monthly, therefore:

$$m = 12$$
$$i^{(12)} = 0,23$$

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

Step 2: Substitute values and solve for i

$$1 + i = \left(1 + \frac{0,23}{12}\right)^{12}$$
$$\therefore i = \left(1 + \frac{0,23}{12}\right)^{12} - 1$$
$$= 25,59\%$$

Step 3: Write the final answer

The effective interest rate is 25,59% per annum.

Worked example 16: Nominal and effective interest rates

QUESTION

Determine the nominal interest rate compounded quarterly if the effective interest rate is 9% per annum (correct to two decimal places).

SOLUTION

Step 1: Write down the known variables

Interest is being added quarterly, therefore:

$$m = 4$$

$$i = 0,09$$

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

Step 2: Substitute values and solve for $i^{(m)}$

$$1 + 0,09 = \left(1 + \frac{i^{(4)}}{4}\right)^4$$

$$\sqrt[4]{1,09} = 1 + \frac{i^{(4)}}{4}$$

$$\sqrt[4]{1,09} - 1 = \frac{i^{(4)}}{4}$$

$$4 \left(\sqrt[4]{1,09} - 1\right) = i^{(4)}$$

$$\therefore i^{(4)} = 8,71\%$$

Step 3: Write the final answer

The nominal interest rate is 8,71% p.a. compounded quarterly.

Exercise 9 – 6: Nominal and effect interest rates

1. Determine the effective annual interest rate if the nominal interest rate is:

- a) 12% p.a. compounded quarterly.
- b) 14,5% p.a. compounded weekly.
- c) 20% p.a. compounded daily.

2. Consider the following:

- 16,8% p.a. compounded annually.
 - 16,4% p.a. compounded monthly.
 - 16,5% p.a. compounded quarterly.
- a) Determine the effective annual interest rate of each of the nominal rates listed above.
- b) Which is the best interest rate for an investment?
- c) Which is the best interest rate for a loan?

3. Calculate the effective annual interest rate equivalent to a nominal interest rate of 8,75% p.a. compounded monthly.
4. Cebela is quoted a nominal interest rate of 9,15% per annum compounded every four months on her investment of R 85 000. Calculate the effective rate per annum.
5. Determine which of the following would be the better agreement for paying back a student loan:
 - a) 9,1% p.a. compounded quarterly.
 - b) 9% p.a. compounded monthly.
 - c) 9,3% p.a. compounded half-yearly.
6. Miranda invests R 8000 for 5 years for her son's study fund. Determine how much money she will have at the end of the period and the effective annual interest rate if the nominal interest of 6% is compounded:

	Calculation	Accumulated amount	Effective annual interest rate
yearly			
half-yearly			
quarterly			
monthly			

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 5. 2393 6. 2394



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9.5 Summary

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- Simple interest: $A = P(1 + in)$
- Compound interest: $A = P(1 + i)^n$
- Simple depreciation: $A = P(1 - in)$
- Compound depreciation: $A = P(1 - i)^n$
- Nominal and effective annual interest rates: $1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$

Exercise 9 – 7: End of chapter exercises

- Thabang buys a Mercedes worth R 385 000 in 2007. What will the value of the Mercedes be at the end of 2013 if:
 - the car depreciates at 6% p.a. straight-line depreciation.
 - the car depreciates at 6% p.a. reducing-balance depreciation.
- Greg enters into a 5-year hire-purchase agreement to buy a computer for R 8900. The interest rate is quoted as 11% per annum based on simple interest. Calculate the required monthly payment for this contract.
- A computer is purchased for R 16 000. It depreciates at 15% per annum.
 - Determine the book value of the computer after 3 years if depreciation is calculated according to the straight-line method.
 - Find the rate according to the reducing-balance method that would yield, after 3 years, the same book value as calculated in the previous question.
- Maggie invests R 12 500 for 5 years at 12% per annum compounded monthly for the first 2 years and 14% per annum compounded semi-annually for the next 3 years. How much will Maggie receive in total after 5 years?
- Tintin invests R 120 000. He is quoted a nominal interest rate of 7,2% per annum compounded monthly.
 - Calculate the effective rate per annum (correct to two decimal places).
 - Use the effective rate to calculate the value of Tintin's investment if he invested the money for 3 years.
 - Suppose Tintin invests his money for a total period of 4 years, but after 18 months makes a withdrawal of R 20 000, how much will he receive at the end of the 4 years?
- Ntombi opens accounts at a number of clothing stores and spends freely. She gets herself into terrible debt and she cannot pay off her accounts. She owes Fashion World R 5000 and the shop agrees to let her pay the bill at a nominal interest rate of 24% compounded monthly.
 - How much money will she owe Fashion World after two years?
 - What is the effective rate of interest that Fashion World is charging her?
- John invests R 30 000 in the bank for a period of 18 months. Calculate how much money he will have at the end of the period and the effective annual interest rate if the nominal interest of 8% is compounded:

	Calculation	Accumulated amount	Effective annual interest rate
yearly			
half-yearly			
quarterly			
monthly			
daily			

8. Convert an effective annual interest rate of 11,6% p.a. to a nominal interest rate compounded:
- a) half-yearly
 - b) quarterly
 - c) monthly
9. Joseph must sell his plot on the West Coast and he needs to get R 300 000 on the sale of the land. If the estate agent charges him 7% commission on the selling price, what must the buyer pay for the plot?
10. Mrs. Brown retired and received a lump sum of R 200 000. She deposited the money in a fixed deposit savings account for 6 years. At the end of the 6 years the value of the investment was R 265 000. If the interest on her investment was compounded monthly, determine:
- a) the nominal interest rate per annum
 - b) the effective annual interest rate
11. R 145 000 is invested in an account which offers interest at 9% p.a. compounded half-yearly for the first 2 years. Then the interest rate changes to 4% p.a. compounded quarterly. Four years after the initial investment, R 20 000 is withdrawn. 6 years after the initial investment, a deposit of R 15 000 is made. Determine the balance of the account at the end of 8 years.

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1. 2396 2. 2397 3. 2398 4. 2399 5. 239B 6. 239C
7. 239D 8. 239F 9. 239G 10. 239H 11. 239J



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